UNIT I | B&W PRINT EDITION | PASS KIT INSIDE | ALL QB-3 QUESTIONS

Diodes and Power Supply Circuits

Master Notes — semiconductors and the P-N junction · diode circuits · rectifiers · filters · Zener regulator · LEDs and applications

Built from: Question Bank-3 of Er. Sanjay Kumar Maurya (M), Er. Awanish Kumar Shukla (A) and Er. Sonam Singh (N); Unit-1 notes (semiconductors; filters, Zener, LED); class notebook.

Question ids: M-L5 = Maurya, long Q5 · A-S12 = Shukla, short Q12 · N-L21 = Sonam Singh, long Q21. Crosswords not covered.

Numbers: every numerical answer computed and checked by script. Si 0.7 V, Ge 0.3 V, GaAs 1.2 V; room temperature = 27 °C = 300 K unless stated.

Exam soon and this is new to you? Start at the Pass Kit, page 5

How to Read This Document

Each kind of box is told apart by **border style, shading and a text tag** instead of colour, so the notes print cheaply in black and white. Learn the code once; it is the same on every page.

▣ DEFINITION

Solid border, grey fill. Exam-ready, quotable one-liners.

★ KEY POINT

Thick left bar only. The single most exam-likely fact in a paragraph.

CASE STUDY

Double border. A named real-world example, citable for extra marks.

⚠ MISCONCEPTION

Dashed border. A common trap, stated then corrected.

№ HIGH-YIELD NUMBER

Solid grey fill. Collected again in one cheat-sheet near the end.

✎ MEMORY AID

Double left bar. Used sparingly, only where a "why" doesn't already make it click.

EXAM FORMAT

Reversed tag. A reusable skeleton for structuring a written answer.

╳ EXTRA — BEYOND SYLLABUS

Dotted border. Not examinable alone; skip under time pressure.

▸ RETRIEVE

Thick top bar. Close the notes and answer from memory *first*; the answers are at the end of the chapter. Tags: RECALL · TELL APART · APPLY · TRANSFER.

⇄ TELL APART

A two-column table for look-alikes, with the one difference that matters in bold.

A	B
…	…

▭ FILL IN

Thick double left bar. Gaps or empty cells to complete from memory; answers sit in the chapter key.

✎ DRAW IT

Dashed border, dotted grid. Sketch it yourself: drawing it helps you recall it.

Diagrams: dark box = the central idea · bold grey = main branches or steps · white = detail · dashed empty box = fill it in. A clean page feels easy to read, which is not the same as knowing it, so close the notes and test yourself.

Contents

PASS KIT — START HERE 5

1 Must know 6

2 Sums 11

3 Extra marks 14

4 Last look (and the exam room) 16

PART B — MASTER NOTES 18

1 · SEMICONDUCTORS AND THE P-N JUNCTION DIODE 18

1.1 Semiconductors and energy bands 18

1.2 Intrinsic and extrinsic semiconductors, doping 19

1.3 The depletion layer and barrier potential 20

1.4 Forward and reverse bias: how the diode works 21

1.5 The V-I characteristic, knee and breakdown voltage 22

1.6 The diode current equation (Shockley) 23

1.7 Effect of temperature on the V-I characteristic 26

1.8 Ideal diode vs real diode; equivalent models 27

1.9 Diode as a switch 27

1.10 Static and dynamic resistance; transition and diffusion capacitance 28

2 · SOLVING DIODE CIRCUITS 29

2.1 Diode circuits with DC sources 29

3 · RECTIFIERS: HALF-WAVE, CENTRE-TAP AND BRIDGE 33

3.1 Half-wave rectifier (HWR) 33

3.2 Deriving the HWR formulas 33

3.3 Rectifier figures of merit (definitions) 35

3.4 Centre-tap full-wave rectifier 35

3.5 Full-wave bridge rectifier 36

3.6 Comparing the three rectifiers 37

3.7 Rectifier sums with a transformer 38

3.8 Sketching outputs of unusual bridge circuits 41

4 · FILTERS AND THE REGULATED DC POWER SUPPLY 44

4.1 Filters and the capacitor filter 44

4.2 Ripple with a capacitor filter 45
4.3 The regulated DC power supply 46

5 · THE ZENER DIODE AND THE ZENER REGULATOR 48

5.1 Breakdown mechanisms: Zener and avalanche 48
5.2 Zener ratings and terms 49
5.3 The Zener as a shunt voltage regulator 50
5.4 Solving Zener networks 51

6 · LEDS AND REAL-WORLD POWER CIRCUITS 55

6.1 The light-emitting diode (LED) 55
6.2 The current-limiting resistor 56
6.3 USB chargers and Arduino / Raspberry Pi supplies 57
6.4 LED indicators and protection diodes 58

BACK MATTER 60

Formula sheet 60
High-yield numbers 60
Glossary 61
Exam-answer templates 62
Mixed quick-fire questions 63
Question-bank coverage map 64
How to revise 65
One-page summary 67

► Start here

Exam soon and **Diodes and Power Supply Circuits** is new to you? You do not need to read the notes. Do this:

1. **Must know** (page **6**). For each card: read the question, try to answer, then read the answer out loud. Do every card.
2. **Go round again.** Cover the answer column with a sheet of paper. Say the answer, uncover, tick a box if you got it. Keep going round until nearly every card has a tick.
3. **Sums** (page **11**). Read the worked example, redo it with the steps covered, then try your turn.
4. **Extra marks** (page **14**), only if you still have time. Same way.
5. **Just before the exam:** read **Last look** (page **16**) twice. It ends with what to do in the exam room.

A card looks like this. One box per time round.

QUESTION	ANSWER (COVER THIS)	GOT IT?
What is a node?	A point where two or more parts of a circuit join.	☐ ☐

Tick only if you said it before you looked. “Looks familiar” is not a tick.
Night before? Sleep is part of revising. Stop and sleep rather than do more cards.

1 Must know

Every card, in order. **First time through:** read the question, try an answer in your head, then read the answer **out loud**. **Then go round again** with the answer column covered: say it, uncover, tick if you were right.

QUESTION	ANSWER (COVER THIS)	GOT IT?
SEMICONDUCTORS **Define semiconductor. Two examples.**	Conductivity between conductor and insulator, controlled by temperature, light and impurities. **Si, Ge**.	☐☐
Intrinsic vs extrinsic semiconductor?	Intrinsic: pure, n = p. Extrinsic: **doped** with impurity, majority and minority carriers.	☐☐
Doping, and why needed?	Adding a tiny controlled impurity to raise conductivity and make P or N regions.	☐☐
Trivalent and pentavalent impurities?	Trivalent (P-type): **B, Al, Ga, In**. Pentavalent (N-type): **P, As, Sb**.	☐☐
Band gap of Si and Ge?	**Si 1.12 eV, Ge 0.72 eV**. Insulator > 5 eV; conductor: bands overlap.	☐☐
What is a hole?	The vacancy left when an electron leaves a covalent bond; acts as a positive charge.	☐☐
P-N JUNCTION **What is the depletion layer? How does it form?**	① Region at the junction with **no free carriers**, only fixed ions. ② Electrons diffuse N → P, holes P → N, and recombine. ③ Fixed ions left: + on N side, − on P side. ④ Their field makes the **barrier** that stops more diffusion. ⑤ Called **space-charge region** because of the uncovered ion charge.	☐☐
Barrier potential of Si and Ge?	**Si 0.7 V, Ge 0.3 V**.	☐☐
Forward bias: connection and effect?	P to +, N to −. Depletion layer **narrows**, barrier falls, large current after 0.7 V.	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
Reverse bias: connection and effect?	P to −, N to +. Depletion layer **widens**, only tiny I_S until breakdown.	☐☐
Why does the depletion layer narrow in forward bias?	Applied voltage pushes majority carriers into the junction; they neutralise fixed ions, so the layer shrinks.	☐☐
Knee voltage? Breakdown voltage?	Knee (cut-in, threshold): forward V where real conduction starts (Si 0.7, Ge 0.3). Breakdown: reverse V where current rises sharply.	☐☐
Draw the diode V-I characteristic: what to mark?	① Forward: flat, then steep rise at **0.7 V** knee. ② Reverse: tiny constant $\mathbf{I_S}$ (µA). ③ Sharp rise at $\mathbf{V_{BR}}$. ④ Label axes I_F mA, I_R µA.	☐☐
Reverse saturation current: carried by? limited by?	**Minority carriers**; their number depends on **temperature**, not voltage, so the current saturates.	☐☐
DIODE EQUATION & CIRCUITS **Diode equation?**	$\mathbf{I_D = I_S(e^{V_D/nV_T} - 1)}$, $V_T = kT/q$.	☐☐
V_T at 27 °C?	T = 300 K: $V_T = 1.38\times10^{-23}\times300/1.6\times10^{-19}$ = **25.88 mV**.	☐☐
P-N JUNCTION **Effect of temperature on V-I curve?**	Forward curve shifts **left** (−2.5 mV/°C); I_S **doubles every 10 °C**.	☐☐
Ideal diode?	Forward: 0 V, 0 Ω (short). Reverse: open, 0 A. No breakdown.	☐☐
Three diode models?	Ideal (switch) · simplified (0.7 V battery) · piecewise-linear (0.7 V + r_{av}).	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
DIODE EQUATION & CIRCUITS **Solving a diode circuit: steps?**	(1) Decide ON/OFF from the arrow direction. (2) ON: replace by 0.7 V (Si), 0.3 (Ge), 1.2 (GaAs). OFF: open. (3) KVL + Ohm's law. (4) Check current flows along the arrow.	☐☐
RECTIFIERS **Rectifier? Half-wave working in one line?**	AC → pulsating DC. HWR: one diode, conducts only on + halves.	☐☐
V_{dc}, I_{rms} of HWR and FWR?	HWR: V_m/π, $I_m/2$. FWR: $\mathbf{2V_m/\pi}$, $I_m/\sqrt{2}$.	☐☐
Ripple factor and efficiency, HWR vs FWR?	HWR: γ = **1.21**, η = **40.6%**. FWR: γ = **0.482**, η = **81.2%**.	☐☐
PIV of HWR, centre-tap, bridge?	V_m · $\mathbf{2V_m}$ · V_m.	☐☐
Define ripple factor, efficiency, TUF.	γ = AC rms / DC. η = P_{dc}/P_{ac}. TUF = P_{dc}/VA rating of secondary.	☐☐
Bridge rectifier working?	(1) Four diodes, AC on one diagonal, load on the other. (2) + half: D1, D3 ON. (3) − half: D2, D4 ON. (4) Load current same direction both halves: full-wave, PIV V_m.	☐☐
Centre-tap rectifier working?	(1) CT transformer + 2 diodes. (2) + half: D1 ON; − half: D2 ON. (3) Same load direction: full-wave. (4) PIV = $2V_m$; η 81.2%, γ 0.482.	☐☐
FILTERS & SUPPLY **Filter? Where are C and L connected?**	Removes ripple. **C in parallel** with load, **L in series**.	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
How does a capacitor filter work?	① Charges to V_m at each peak. ② Diodes go OFF; C discharges slowly into R_L. ③ Next peak tops it up: small ripple left. ④ X_C low for AC, open for DC.	☐☐
Regulated DC supply blocks?	**Transformer → rectifier → filter → regulator** → load.	☐☐
ZENER **Zener vs avalanche breakdown?**	Zener: heavy doping, thin layer, tunnelling, < 5 V, **negative** temp coeff. Avalanche: light doping, collisions, > 6 V, **positive**.	☐☐
Why is avalanche temp coefficient positive, Zener negative?	Heat → more lattice collisions so carriers need more V (avalanche ↑); heat helps electrons tunnel (Zener ↓).	☐☐
Zener as voltage regulator?	① Zener **reverse biased**, in **parallel** with load; R_S in series. ② $I_S = I_Z + I_L$. ③ V_{in} or load changes → I_Z changes, V_L stays V_Z. ④ Works while $I_{ZK} < I_Z < I_{ZM}$.	☐☐
Zener sum: first step?	**ON/OFF test**: remove Zener, $V = V_{in}R_L/(R + R_L)$. ON only if $V \geq V_Z$.	☐☐
LED & APPLICATIONS **LED: what and how?**	P-N diode that emits light in forward bias by **electroluminescence**: electron-hole recombination releases photons.	☐☐
LED series resistor formula? 5 V, V_F = 2 V, 10 mA?	**$R = (V_S - V_F)/I_F$** = 3/0.01 = **300 Ω**.	☐☐
Why a resistor in series with an LED?	LED resistance is tiny past V_F; current would rise too much and burn it. R limits current.	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
Rectifier in a USB charger: function?	Converts AC to DC for charging; output then filtered and regulated to 5 V.	☐☐
Arduino Uno operating voltage?	**5 V** (input 7-12 V via regulator); 3.3 V rail also provided.	☐☐

End of Must know → back to the first card and go round again: cover, say, check, tick.

2 Sums

Watch one, then do one. Read the worked example, then cover its steps and redo it on paper. Then try your turn.

WORKED EXAMPLE — ZENER: IS IT ON, AND WHAT ARE THE CURRENTS?

GIVEN 16 V supply, R = 1 kΩ, V_Z = 10 V, R_L = 3 kΩ.

FIND V_L, I_Z, P_Z.

1. Remove the Zener: 16 × 3/(1 + 3) → **12 V, above 10 V, so ON**
2. Load voltage → **$V_L = V_Z$ = 10 V**
3. Resistor current: (16 − 10)/1 k → **6 mA**
4. Load current: 10/3 k → **3.33 mA**
5. Zener current: 6 − 3.33 → **2.67 mA**
6. Power: 10 × 2.67 mA → **26.7 mW**

ANSWER **V_L = 10 V, I_Z = 2.67 mA, P_Z = 26.7 mW**

WORKED EXAMPLE — RECTIFIER THROUGH A TRANSFORMER

GIVEN 230 V AC, 10 : 1 transformer, ideal half-wave rectifier.

FIND V_{dc} and PIV.

1. Secondary RMS: 230/10 → **23 V**
2. Peak: √2 × 23 → **32.53 V**
3. DC output: V_m/π → **10.35 V**
4. PIV = V_m → **32.53 V**

ANSWER **V_{dc} = 10.35 V, PIV = 32.53 V**

Your turn: solve on paper, then check

QUESTION	ANSWER (COVER THIS)	GOT IT?
RECTIFIERS **230 V, 10 : 1, ideal HWR. V_{dc} and PIV?**	$V_m = \sqrt{2}\times 23 = 32.53$ V; $V_{dc} = V_m/\pi =$ **10.35 V**; PIV = **32.53 V**.	☐☐
FILTERS & SUPPLY **Bridge, 220 V, 10 : 1, 470 µF, 1 kΩ, 50 Hz. V_m, ripple, V_{dc}?**	$V_m = 31.11$ V; $V_r = V_m/(2fCR) =$ **0.662 V**; $V_{dc} = V_m - V_r/2 =$ **30.78 V**.	☐☐
RECTIFIERS **r_f 20 Ω, 50 sinωt, R_L 800 Ω, HWR. I_m, I_{dc}, η?**	$I_m = 50/820 = 60.98$ mA; $I_{dc} = 19.41$ mA; $I_{rms} = 30.49$ mA; η = **39.5%**.	☐☐
Bridge 120 V rms, 1 kΩ, Si. V_{dc}, PIV, I_{max}?	$V_{L(peak)} = 169.71 - 1.4 = 168.31$ V; $V_{dc} =$ **107.1 V**; PIV ≥ 169 V; I = 168.3 mA; $P_D = 117.8$ mW.	☐☐
ZENER **Zener: 16 V, R 1 k, V_Z 10 V, R_L = 3 kΩ. I_Z, P_Z?**	Test: 12 V > 10 so ON. $I_R = 6$ mA, $I_L = 3.33$ mA, $I_Z =$ **2.67 mA**, $P_Z =$ **26.7 mW**. (With 1.2 k: OFF, $V_L = 8.73$ V.)	☐☐
80-120 V, 5 k, V_Z 50 V, R_L 10 k. I_Z max/min?	$I_L = 5$ mA. I = 6 mA to 14 mA. $I_Z =$ **1 mA to 9 mA**.	☐☐
R 250 Ω, V_Z 20 V, I_{Zmax} 50 mA, R_L 2 k. Range of V_s?	$I_L = 10$ mA. Min: 20 + 250×0.01 = **22.5 V**. Max: 20 + 250×0.06 = **35 V**.	☐☐
20 V, 220 Ω, V_Z 10 V, R_L 180 Ω. V_L, I_Z?	Test: 20×180/400 = 9 V < 10: OFF. **V_L = 9 V, I_L = I_R = 50 mA, I_Z = 0**.	☐☐
DIODE EQUATION & CIRCUITS **I_s = 40 nA, n = 2, 0.5 V, 20 °C. I_D?**	$V_T = 25.27$ mV; exponent = 9.89; $I_D =$ **0.79 mA**.	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
RECTIFIERS **230 V, 60 Hz, 5 : 1 CT, R_L 900 Ω, r 100 Ω. V_{dc}, PIV?**	Each half 23 V rms, V_m = 32.53 V; I_m = 32.53 mA; $I_{dc} = 2I_m/\pi$ = 20.71 mA; V_{dc} = **18.64 V**; P_{dc} = 0.386 W; PIV = $2V_m$ = **65.1 V**.	☐☐
HWR gives 50 V DC to 800 Ω, r_f 25 Ω. AC input?	I_{dc} = 62.5 mA; $I_m = \pi I_{dc}$ = 0.1963 A; $V_m = I_m \times 825$ = 162.0 V; V_{rms} = **114.5 V**.	☐☐
230 V, 4 : 1, ideal bridge, 200 Ω. V_{dc}, PIV, f_{out}?	V_s = 57.5 V, V_m = 81.32 V; V_{dc} = **51.77 V**; PIV = 81.32 V; f_{out} = **100 Hz**.	☐☐
ZENER **V_Z 10 V; V_{in} 13-16 V; I_L 10-85 mA; I_{Zmin} 15 mA. R?**	Worst case: R = (13 − 10)/(85 + 15 mA) = **30 Ω**.	☐☐

3 Extra marks

Only once most Must-know cards have a tick. Same way: read, say, then cover and check.

QUESTION	ANSWER (COVER THIS)	GOT IT?
P-N JUNCTION **Why is it called space-charge region?**	It holds uncovered fixed ions, a layer of charge on each side.	☐☐
Static vs dynamic resistance?	Static R = V/I at a point. Dynamic $r_d = \Delta V/\Delta I \approx 26$ mV/I_D.	☐☐
Transition vs diffusion capacitance?	Transition: **reverse** bias, depletion layer, pF. Diffusion: **forward** bias, stored charge, larger.	☐☐
Diode as a switch?	Forward = closed (ON, 0.7 V). Reverse = open (OFF). Used in rectifiers, protection, logic.	☐☐
SEMICONDUCTORS **Why Si preferred over Ge?**	Smaller I_s, higher temperature and PIV, cheap and plentiful.	☐☐
Why semiconductors have negative temp coefficient?	Heat makes more electron-hole pairs, so resistance falls.	☐☐
Donor level position?	Just **below the conduction band** (about 0.01-0.05 eV).	☐☐
Recombination?	Free electron falls into a hole; both vanish, energy released.	☐☐
RECTIFIERS **Voltage regulation of a rectifier? Ideal value?**	$(V_{NL} - V_{FL})/V_{FL} \times 100$; ideal **0%**. HWR: $r_f/R_L \times 100$, worse as r_f rises.	☐☐
Bridge over centre-tap: two advantages?	No centre tap needed; PIV only V_m; higher TUF (0.812).	☐☐
Disadvantages and uses of HWR?	High ripple, 40.6%, low TUF, core saturation. Uses: small chargers, detectors.	☐☐

QUESTION	ANSWER (COVER THIS)	GOT IT?
FILTERS & SUPPLY **Four filter types?**	C, L, LC (L-section), π (C-L-C).	☐☐
ZENER **Derating factor?**	Reduction of max power rating per °C above 25 °C (mW/°C).	☐☐
Knee Zener current?	Minimum reverse current I_{ZK} needed to stay in breakdown (regulating).	☐☐
4.5 V, 1.5 W Zener: max current?	I = P/V = 1.5/4.5 = **333 mA**.	☐☐
Zener applications?	Regulator, reference voltage, over-voltage / surge protection, clipper.	☐☐
Zener regulator: 2 advantages, 2 limits?	Simple, cheap. Wastes power; only small load current.	☐☐
LED & APPLICATIONS **P-N diode vs LED?**	LED: GaAs/GaN, emits light, V_F 1.8-3.3 V, low reverse rating. Diode: Si, heat, 0.7 V.	☐☐
LED precautions?	Series resistor, right polarity (< 5 V reverse), rated current, no overheating.	☐☐
Why Raspberry Pi needs stable supply?	Dips cause resets, SD corruption, throttling, USB/Wi-Fi drop-outs.	☐☐
Reverse-polarity protection by a diode?	Series diode conducts only when supply is the right way round; reversed, it blocks.	☐☐
Protection diodes on Arduino pins?	Clamp diodes to V_{CC} and GND steer over-voltage and ESD away from the chip.	☐☐
LED indicator examples?	Charger (charging/full), router (activity), TV standby, laptop power.	☐☐

4 Last look

Just before the exam: read this straight through, twice. Nothing new now.

Semiconductors

Si 1.12 eV, Ge 0.72 eV. N-type: P/As/Sb, electrons. P-type: B/Al/Ga/In, holes.

n = p in intrinsic

P-N junction

Depletion layer = no free carriers. FB narrows, RB widens. Barrier Si 0.7, Ge 0.3 V.

I_s doubles / 10 °C

Diode equation & circuits

$I = I_s(e^{V/nV_T} - 1)$; $V_T = kT/q = 25.9$ mV at 300 K.

Diode ON → 0.7 V source

Rectifiers

HWR: V_m/π, 1.21, 40.6%, PIV V_m. FWR: $2V_m/\pi$, 0.482, 81.2%.

PIV: CT $2V_m$, bridge V_m

Filters & supply

C across load, L in series. $V_r = V_m/(2fCR_L)$.

Transformer → rectifier → filter → regulator

Zener

Zener: heavy doping, tunnelling, − temp coeff. Avalanche: collisions, +.

Test first; $I_S = I_Z + I_L$

LED & applications

Electroluminescence. $R = (V_S - V_F)/I_F$.

5 V, 2 V, 10 mA → 300 Ω

IF YOU REMEMBER ONE THING

Diode = one-way valve. 0.7 V Si. HWR 40.6%, FWR 81.2%. Zener holds V_Z in reverse. LED needs a resistor.

Then, in the exam

1. **Nervous?** That is your body getting ready. Breathe out slowly, then start.
2. **Read every question first.** Then do the easiest ones first.
3. **Never leave a blank.** Write what you know: the definition, the main points, an example. Part marks are real marks.

4 **Multiple choice:** cross out the two worst options, then pick. Change an answer only if you have a reason.

5 **Last 5 minutes:** fill any blanks; check units and signs.

1 Semiconductors and the P-N junction diode

UNDERSTAND *why a P-N junction conducts one way only* · **RECALL EXACTLY** *the definitions, barrier values and the diode equation* · **BE ABLE TO** *draw the V-I curve and solve diode-equation sums.*

1.1 Semiconductors and energy bands

QB: M-S5, M-S7, M-S8, A-S9, A-S7

▣ **DEFINITION**

A **semiconductor** is a material whose electrical conductivity lies between that of a conductor and an insulator, and can be controlled by temperature, light, electric field and added impurities. Examples: **silicon (Si), germanium (Ge)**, gallium arsenide (GaAs).

Electrons in a solid can only have energies inside allowed **bands**. The **valence band** holds bound electrons; the **conduction band** holds free electrons that carry current. The **forbidden energy gap E_g** between them decides the class of material.

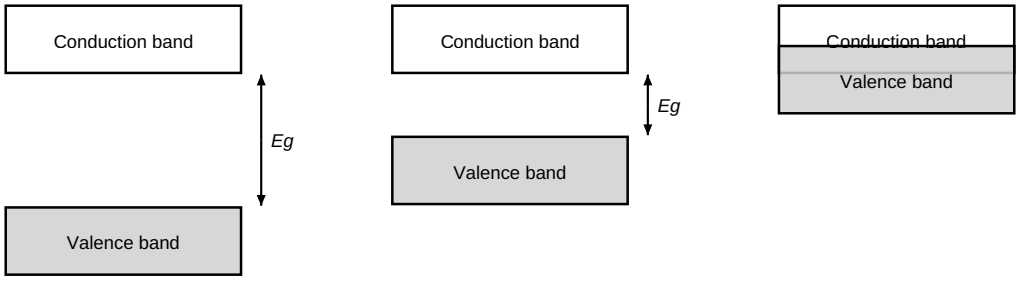


Fig 1.2 — Energy band diagrams

№ **HIGH-YIELD NUMBER**

Band gap at room temperature: **Si 1.12 eV**, **Ge 0.72 eV** (Ge is often quoted as 0.7 eV). Insulator: more than about 5 eV.

▣ **DEFINITIONS**

Energy band gap: the minimum energy an electron needs to jump from the valence band to the conduction band. **Recombination**: a free electron falls back into a hole; both carriers disappear and the energy is given out (as heat in Si, as light in an LED).

★ KEY POINT

Why semiconductors have a NEGATIVE temperature coefficient of resistance: heat breaks covalent bonds and creates more electron-hole pairs, so conductivity rises and resistance falls as temperature rises. **Why a pure semiconductor is an insulator near 0 K:** there is no thermal energy to break bonds, so there are no free carriers.

1.2 Intrinsic and extrinsic semiconductors, doping

QB: M-S1, A-S1, N-S9, A-S4, A-S5, A-S6, N-S2, A-S8

▣ DEFINITION

Intrinsic semiconductor: pure Si or Ge with no added impurity; free electrons = holes ($n = p = n_i$).

Extrinsic semiconductor: one whose conductivity has been raised by adding a controlled impurity (doping).

▣ DEFINITION

Doping: adding a tiny, controlled amount of impurity to a pure semiconductor to increase its conductivity and decide which carrier is in the majority. **Why required:** intrinsic conductivity is far too low, and devices need separate P and N regions.

▣ DEFINITION

Hole: the vacancy left in a covalent bond when an electron leaves it. It behaves like a positive charge ($+1.6 \times 10^{-19}$ C) and moves when a neighbouring electron fills it.

Table 1.1 — N-type versus P-type

	N-type	P-type
Impurity	Pentavalent (5 valence e^-): **P, As, Sb**	Trivalent (3 valence e^-): **B, Al, Ga, In**
Impurity called	Donor (gives an electron)	Acceptor (accepts an electron)
Majority carrier	Electrons	Holes
Minority carrier	Holes	Electrons
Extra energy level	Donor level just **below the conduction band**	Acceptor level just above the valence band

Donor level (A-S8): draw the band diagram of Fig 1.2 for a semiconductor and add a dashed line a little *below* the conduction band, labelled “donor level”. Its electrons need only about 0.01 eV (Ge) or 0.05 eV (Si) to enter the conduction band, so at room temperature almost all of them are free.

Table 1.2 — Intrinsic versus extrinsic (5 rows are enough for a 2-mark answer)

Basis	Intrinsic	Extrinsic
Purity	Pure Si / Ge	Impurity deliberately added

Basis	Intrinsic	Extrinsic
Carriers	$n = p = n_i$	Majority and minority carriers, unequal
Conductivity	Low	High, set by doping
Depends on	Temperature only	Doping level and temperature
Types	None	N-type and P-type

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

1.2-1 RECALL Name two trivalent and two pentavalent impurities.

1.2-2 TELL APART Which carrier is in the majority in P-type, and why?

1.2-3 APPLY Why does resistance of pure Si fall when it is heated?

1.3 The depletion layer and barrier potential

QB: M-L3, A-L1, N-L1, N-S1, N-S3, N-S5

▣ **DEFINITION (N-S1: WHAT IS A DIODE?)**

A **P-N junction diode** is a two-terminal device made by forming P-type and N-type regions in one crystal. It conducts easily in one direction and blocks current in the other.

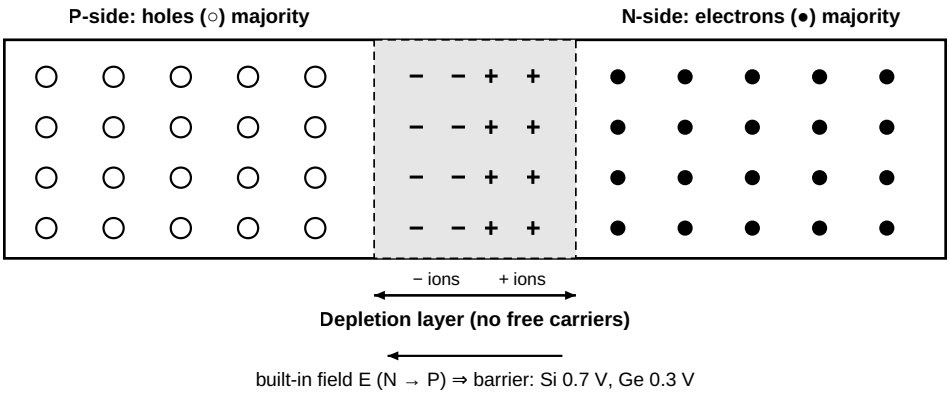


Fig 1.1 — P-N junction at no bias: depletion layer and barrier

Formation, in five steps (write these for any “explain its formation” question):

1. Electrons **diffuse** from the N-side (where there are many) to the P-side; holes diffuse from P to N.
2. Near the junction they **recombine** and disappear.
3. This leaves **fixed ions**: positive donor ions on the N-side, negative acceptor ions on the P-side.
4. The thin region near the junction now has **no free carriers**: this is the **depletion layer**.

5 The ions set up an electric field (N to P) and a **barrier potential** that stops further diffusion: equilibrium.

▣ **DEFINITION**

Depletion layer: the region on both sides of a P-N junction that is emptied of mobile charge carriers and contains only immobile ions. It is called the **space-charge region** because the uncovered fixed ions form a layer of charge (space charge) on each side.

▣ **DEFINITION**

Barrier potential: the potential difference across the depletion layer that opposes diffusion of majority carriers. **Si ≈ 0.7 V, Ge ≈ 0.3 V** at room temperature.

In reverse bias (M-L3) the battery pulls holes toward its negative terminal and electrons toward its positive terminal, away from the junction. More ions are uncovered, so the depletion layer becomes **wider** and the barrier **higher**. The larger the reverse voltage, the wider the layer.

1.4 Forward and reverse bias: how the diode works

QB: M-L2, A-L3, N-L7, N-S8, M-S3, A-S2, N-S10, A-L2, N-S7

▣ **DEFINITION (N-S8)**

Biasing: applying an external DC voltage across a device to set its operating condition.

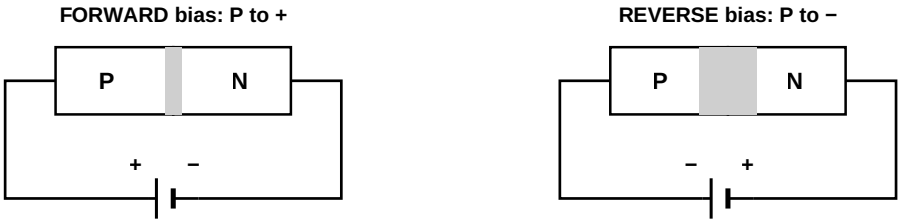


Fig 1.6 — Forward and reverse bias (shaded = depletion layer)

	Forward bias	Reverse bias
LOOK ALIKE BECAUSE	the same diode and the same battery; only the direction is changed	
CONNECTION	P to +, N to −	P to −, N to +
DEPLETION LAYER	Narrows	Widens
BARRIER	Reduced	Increased
CURRENT	Large (mA) once V > 0.7 V	Tiny I_S (nA to µA) until breakdown
RESISTANCE	Low	Very high
THE DIFFERENCE THAT MATTERS	**Battery opposes the barrier: diode ON**	**Battery adds to the barrier: diode OFF**

★ KEY POINT

Why the depletion layer narrows in forward bias (M-S3, A-S2, N-S10): the applied voltage pushes majority carriers (holes from P, electrons from N) toward the junction. They neutralise some of the fixed ions, so the uncovered-ion region shrinks and the barrier falls.

Reverse saturation current (A-L2, N-S7)

▣ DEFINITION

Reverse saturation current I_s: the small current that flows in a reverse-biased diode, carried by **minority carriers** (electrons in P, holes in N), for which the junction field is a downhill path.

► **Reverse current carriers:** minority carriers.

► **What limits their number:** they are made only by thermal generation of electron-hole pairs, so their number depends on **temperature**, not on the applied voltage.

► **Why "saturation":** even a small reverse voltage sweeps all of them across; raising the voltage cannot make more, so the current stays constant (saturates). Si: nA; Ge: µA.

HOW TO WRITE IT IN THE EXAM — EXPLAIN THE WORKING OF A P-N JUNCTION DIODE (M-L2, A-L3, N-L7)

1. Define the diode (one line) and draw its symbol: arrow = anode (P), bar = cathode (N).
2. No bias: diffusion → depletion layer → barrier 0.7 V (Si) / 0.3 V (Ge). Draw Fig 1.1.
3. Forward bias: circuit + 3 effects (layer narrows, barrier falls, large current after 0.7 V).
4. Reverse bias: circuit + 3 effects (layer widens, barrier rises, only I_s; breakdown at V_{BR}).
5. Draw the V-I characteristic (Fig 1.3) with knee and breakdown marked.
6. Close: "Hence the diode conducts in one direction only and acts as a one-way switch."

1.5 The V-I characteristic, knee and breakdown voltage

QB: M-L1, M-L4, M-S9, N-S6, N-S4

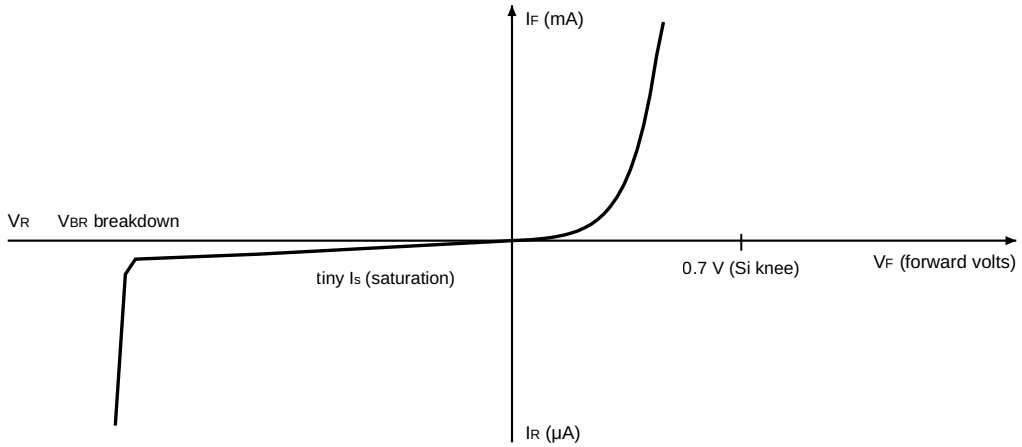


Fig 1.3 — V-I characteristic of a silicon diode (axes not to the same scale)

Forward region: almost no current until the barrier is overcome, then current rises steeply (exponentially). **Reverse region:** a tiny constant I_S; at the breakdown voltage the current rises sharply.

▣ **DEFINITIONS (M-S9, N-S6)**

Knee (cut-in, threshold) voltage: the forward voltage at which the diode starts to conduct significant current; it equals the energy needed to overcome the barrier. **Si 0.7 V, Ge 0.3 V**.
Breakdown voltage V_{BR}: the reverse voltage at which the reverse current rises sharply. An ordinary diode can be destroyed here unless the current is limited.

Applications of diodes (N-S4): rectifiers in power supplies · reverse-polarity protection · clippers and clampers · voltage regulation (Zener) · LED indicators · switching and logic gates.

1.6 The diode current equation (Shockley)

QB: M-L4, M-L8, M-L9, N-L6, A-L7, A-L8, A-L9, M-S6, N-S14

The equation that gives the diode current for any applied voltage. Most sums: find I, find I_S, or find V.

You'll see it as: "write down and explain the junction diode equation" · "find the diode current / I_S / forward voltage" · "determine the thermal voltage at 20 °C"

THE IDEA

1. Convert temperature to kelvin: T = °C + 273.
2. Find the thermal voltage $V_T = kT/q$.
3. Work out the exponent $V_D/(nV_T)$ as a plain number first.
4. Put it in the equation and solve for the unknown.

THE FORMULA

$$I_D = I_s\left(e^{V_D/nV_T} - 1\right), \qquad V_T = \frac{kT}{q}$$

I_s = reverse saturation current · V_D = diode voltage (+ forward) · n = ideality factor (class notes: 1 for Ge, 2 for Si; use the value given) · k = 1.38×10^{-23} J/K · q = 1.6×10^{-19} C

Golden rule: find V_T first, in volts, from the temperature in kelvin; the exponent must be a plain number.

Explaining the equation (M-L4): in forward bias the exponential term dominates, so current grows very fast with voltage; in reverse bias the exponential becomes nearly 0, so $I_D \approx -I_s$.

Solved example

QB M-L9 / N-L6

(a) Find V_T at 20 °C. (b) Same diode: I_s = 40 nA, n = 2, V_D = 0.5 V. Find I_D.

Step 1: temperature in kelvin

$$T = 20 + 273 = 293\ \text{K}$$

Step 2: thermal voltage

$$V_T = \frac{kT}{q} = \frac{1.38 \times 10^{-23} \times 293}{1.6 \times 10^{-19}} = 25.27\ \text{mV}$$

Step 3: exponent

$$\frac{V_D}{nV_T} = \frac{0.5}{2 \times 0.02527} = 9.893$$

Step 4: current

$$I_D = 40 \times 10^{-9}\left(e^{9.893} - 1\right) = 40 \times 10^{-9} \times 19785 = 0.791\ \text{mA}$$

V_T = 25.27 mV · $I_D \approx$ 0.79 mA

Check: a forward-biased diode at 0.5 V gives a current far bigger than I_s, as it should.

HOW TO WRITE IT IN THE EXAM

Write: given → T in K → V_T → exponent → I_D, each with units, and box the answer.

More from the question bank, solved

Solved

QB M-L8

I_D = 8 mA, n = 1, V_D = 0.5 V, T = 25 °C. Find I_s.

Working

Thermal voltage at 298 K:

$$V_T = \frac{1.38 \times 10^{-23} \times 298}{1.6 \times 10^{-19}} = 25.70\ \text{mV}$$

Rearrange the equation for I_s:

$$I_s = \frac{I_D}{e^{V_D/nV_T} - 1} = \frac{8 \times 10^{-3}}{e^{19.45} - 1} = \frac{8 \times 10^{-3}}{2.809 \times 10^{8}}$$

$I_s \approx 2.85 \times 10^{-11}$ A

Solved

QB A-L8

Ge diode, I_S = 0.3 µA, forward bias 0.15 V, room temperature. Find the current.

Working

Room temperature = 27 °C = 300 K, Ge so n = 1:

$$V_T = 25.88\ \text{mV}, \quad \frac{0.15}{0.02588} = 5.797$$

$$I = 0.3 \times 10^{-6}\left(e^{5.797} - 1\right) = 0.3 \times 10^{-6} \times 328.3 = 98.5\ \mu\text{A}$$

I ≈ 0.099 mA

Check: your class notes took V_T = 25 mV, giving 0.121 mA. Either is accepted if you state the V_T you used.

Solved

QB A-L7

Ge diode draws 50 mA at 0.27 V, room temperature. Find I_S (i) now (ii) at 47 °C.

Working

n = 1, V_T = 25.88 mV:

$$I_s = \frac{50 \times 10^{-3}}{e^{0.27/0.02588} - 1} = \frac{50 \times 10^{-3}}{e^{10.43} - 1} = 1.47\ \mu\text{A}$$

I_S doubles every 10 °C; 27 → 47 °C is a 20 °C rise, so $\times 2^2$:

$$I_s(47^\circ\text{C}) = 4 \times 1.47 = 5.88\ \mu\text{A}$$

(i) 1.47 µA (ii) 5.88 µA

Solved

QB A-L9

Si diode, $I_S = 25 \times 10^{-7}$ A. Forward voltage for 10 mA (i) at room temperature (ii) at 47 °C.

Working

Rearrange: $V = nV_T \ln(I/I_S + 1)$, Si so n = 2:

$$V = 2 \times 0.02588 \times \ln\left(\frac{10 \times 10^{-3}}{25 \times 10^{-7}} + 1\right) = 0.05175 \times \ln(4001) = 0.429\ \text{V}$$

At 47 °C: $I_S \times 4 = 1 \times 10^{-5}$ A, and V_T at 320 K = 27.60 mV:

$$V = 2 \times 0.0276 \times \ln(1001) = 0.381\ \text{V}$$

(i) 0.429 V (ii) 0.381 V (less voltage needed when hot)

REMEMBER

1. V_T at 27 °C (300 K) = 25.88 mV ≈ 26 mV (M-S6, N-S14). At 20 °C: 25.27 mV.
2. Always use T in kelvin.
3. If n is not given: n = 1 for Ge, n = 2 for Si (class convention).
4. I_S doubles for every 10 °C rise.

IF THE QUESTION...

- **If** asks for V (not I) → take logs: $V = nV_T \ln(I/I_s + 1)$
- **If** gives a new temperature → recompute V_T AND scale I_s by 2 per 10 °C
- **If** says "room temperature" → use 27 °C = 300 K and say so

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

1.6-1 APPLY Find V_T at 27 °C.

1.6-2 APPLY A diode has I_s = 10 nA. Roughly what current flows at V_D = −5 V?

1.7 Effect of temperature on the V-I characteristic

QB: M-L7, A-L5, N-L2, A-S10

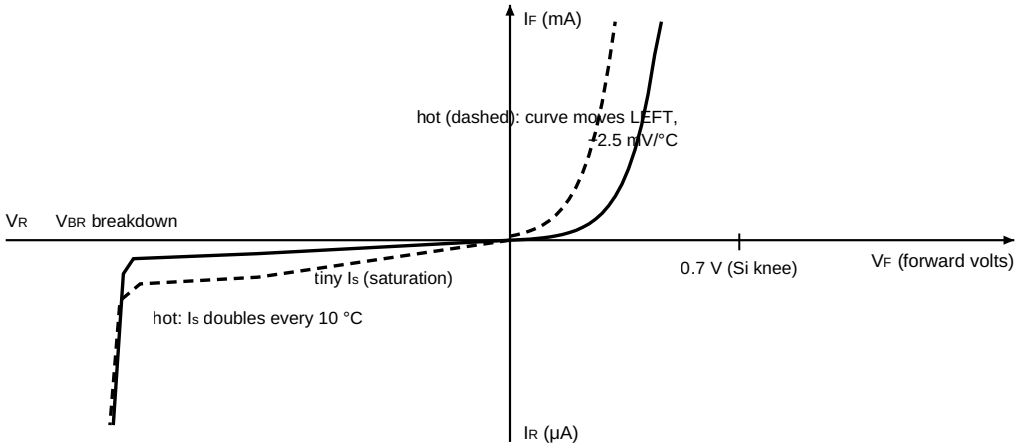


Fig 1.4 — Effect of temperature (dashed = higher temperature)

1. **Forward:** the curve shifts **left**; the knee voltage falls by about **2.5 mV per °C** (Si). The same current flows at a lower voltage.
2. **Reverse:** I_s **doubles for every 10 °C** rise (more thermally made minority carriers).
3. **Breakdown:** V_{BR} changes with temperature (rises for avalanche, falls for Zener breakdown, §5.1).
4. **Why:** heat creates more electron-hole pairs, so more carriers are available at every voltage.

★ **KEY POINT**

Why Si diodes are more popular than Ge (A-S10): much smaller I_s (nA, not µA), work at higher temperature (about 150 °C vs 75 °C) and higher PIV, and silicon is cheap and plentiful.

1.8 Ideal diode vs real diode; equivalent models

QB: N-L3, N-L4, A-L4

Table 1.3 — Ideal versus real diode

Property	Ideal diode	Real Si diode
Forward drop	0 V	≈ 0.7 V
Forward resistance	0 Ω (short)	small, non-zero
Reverse current	0 A	tiny leakage I_S
Reverse resistance	infinite (open)	very high
Breakdown	none	finite V_{BR}
Switching	instant	finite time
Power loss	zero	non-zero

Features of an ideal diode (N-L3): a perfect one-way switch: short circuit when forward biased, open circuit when reverse biased. Its V-I curve is the left graph of Fig 1.5 (vertical line up the current axis, flat line along the negative voltage axis).

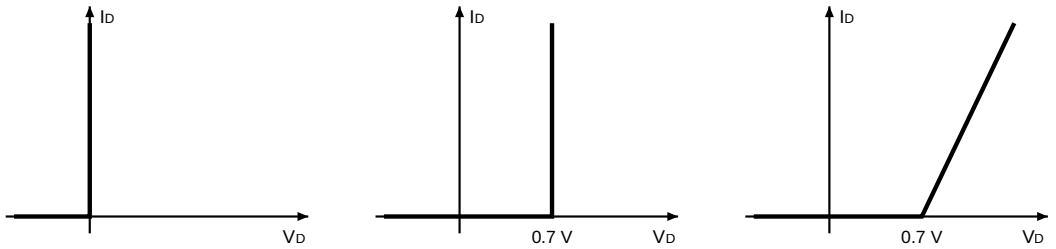


Fig 1.5 — Equivalent models of a P-N junction diode (I-V characteristic of each)

► **Ideal model:** switch only. Use for large voltages (e.g. rectifier with 230 V).

► **Simplified (constant-drop) model:** a 0.7 V battery when ON. Use for most circuit sums (§2).

► **Piecewise-linear model:** 0.7 V battery in series with the average resistance r_{av}. Most accurate of the three.

1.9 Diode as a switch

QB: A-L4

A diode's conduction depends only on the polarity of the voltage across it, so it works as an automatic switch.

► **Forward bias → switch CLOSED (ON):** low resistance, drop 0.7 V (0 V ideal).

► **Reverse bias → switch OPEN (OFF):** very high resistance, only I_s, as long as $V < V_{BR}$.

► **Uses:** rectifiers, reverse-polarity protection, clippers and clampers, logic gates, freewheeling diodes across relay coils.

1.10 Static and dynamic resistance; transition and diffusion capacitance

QB: M-S4, N-L5

▣ **DEFINITIONS**

Static (DC) resistance $R_D = V_D/I_D$ at one operating point. It falls as current rises.

Dynamic (AC) resistance $r_d = \Delta V_D/\Delta I_D$, the inverse slope of the curve at that point; $r_d \approx 26\ \text{mV} / I_D$.

	Transition capacitance C_T	Diffusion capacitance C_D
LOOK ALIKE BECAUSE	both are junction capacitances of the same diode	
BIAS WHERE IT DOMINATES	**Reverse**	**Forward**
CAUSE	Depletion layer = insulator between two charged regions (C = εA/W)	Charge stored by injected minority carriers near the junction
SIZE	Small, pF	Large, nF to µF
VARIES WITH	Falls as reverse voltage rises (layer widens)	Rises with forward current
THE DIFFERENCE THAT MATTERS	**Used in the varactor (voltage-variable capacitor)**	**Limits switching speed**

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

1.10-1 TELL APART Which capacitance dominates in reverse bias, and why?

1.10-2 RECALL Write the diode equation and name each symbol.

1.10-3 NEW SETTING A solar panel charges a battery through a diode. At night the panel gives 0 V. What does the diode do?

ANSWERS — COVER UNTIL YOU HAVE TRIED

RECALL **1.2-1** Trivalent: boron, aluminium (also Ga, In). Pentavalent: phosphorus, arsenic (also Sb).

TELL APART **1.2-2** Holes: each trivalent atom leaves one bond incomplete, creating a hole.

APPLY **1.2-3** Heat creates more electron-hole pairs, so more carriers are available (negative temperature coefficient).

APPLY **1.6-1** 300 K: $V_T = 25.88$ mV.

APPLY **1.6-2** About −10 nA: in reverse bias $I_D \approx -I_s$.

TELL APART **1.10-1** Transition capacitance: the widened depletion layer acts as the dielectric between two charged regions.

RECALL **1.10-2** $I_D = I_s(e^{V_D/nV_T} - 1)$; I_s reverse saturation current, n ideality factor, $V_T = kT/q$ thermal voltage.

NEW SETTING **1.10-3** It is reverse biased by the battery, so it acts as an open switch and stops the battery discharging back through the panel.

2 Solving diode circuits

BE ABLE TO *decide which diodes are ON, replace them by 0.7 V (Si), 0.3 V (Ge) or 1.2 V (GaAs), and find every current and voltage.*

2.1 Diode circuits with DC sources

QB: M-L5, M-L6, A-L6, N-L20

Circuits with one or more diodes and a DC source; you find V_o, I_D, V_R.

You'll see it as: "determine V_o and I_D for the networks" · "determine the indicated currents and voltages"

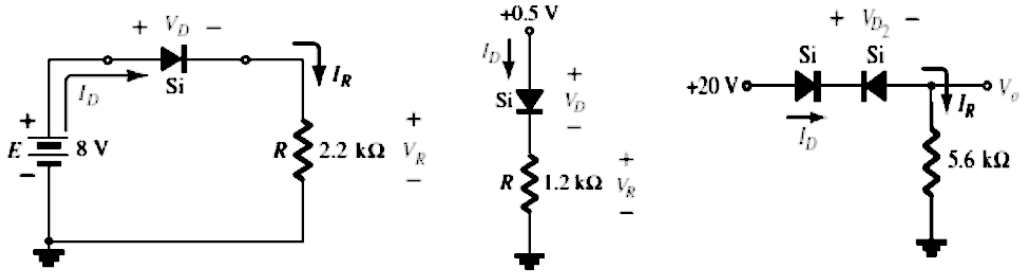
THE IDEA

1. Look at each diode: does the source push current in the direction of its arrow? If yes, it is ON.
2. Replace an ON diode by its drop (Si 0.7 V, Ge 0.3 V, GaAs 1.2 V); an OFF diode by an open circuit.
3. Write KVL around the loop and Ohm's law for the resistor.
4. Check: an ON diode's current must flow in the arrow direction.

THE FORMULA

$$I = \frac{E - V_D}{R}, \qquad V_R = E - V_D$$

Golden rule: two diodes in parallel: the one with the SMALLER drop turns on and fixes the voltage; the other stays off.



QB M-L6 / N-L20 circuits (a), (b), (c)

Solved example

QB M-L6(a) / N-L20(a)

E = 8 V, Si diode in series with R = 2.2 kΩ.

Step 1: state of the diode

The source pushes current along the arrow, so the diode is ON: V_D = 0.7 V.

Step 2: KVL and Ohm

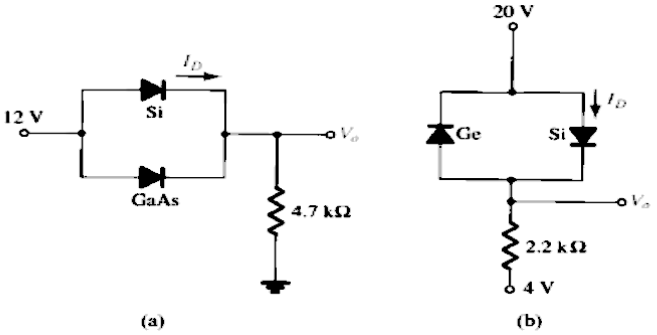
$$V_R = 8 - 0.7 = 7.3 \text{ V}$$
$$I_D = I_R = \frac{7.3}{2.2 \text{ k}\Omega} = 3.32 \text{ mA}$$

$V_D = 0.7$ V, $V_R = 7.3$ V, $I_D = I_R = 3.32$ mA

HOW TO WRITE IT IN THE EXAM

Part	Diode state	Result
(a)	Si ON	$V_R = 7.3$ V, $I = 3.32$ mA
(b)	OFF: 0.5 V < 0.7 V needed	$I_D = 0$, $V_R = 0$, $V_D = 0.5$ V
(c)	OFF: second Si diode is reversed	$I = 0$, $V_o = 0$ V, $V_{D2} = 20$ V (whole supply across the reversed diode)

Hence state each diode, then each value, with units.



QB M-L5 circuits (a), (b)

Solved

QB M-L5

(a) 12 V, Si and GaAs diodes in parallel, 4.7 kΩ. (b) 20 V, Ge (reversed) and Si in parallel, 2.2 kΩ to 4 V.

(a) Parallel diodes

Si needs 0.7 V, GaAs needs 1.2 V. Si turns on first and holds the pair at 0.7 V, so GaAs stays OFF:

$$V_o = 12 - 0.7 = 11.3\ \text{V}, \quad I_D = \frac{11.3}{4.7\ \text{k}\Omega} = 2.40\ \text{mA}$$

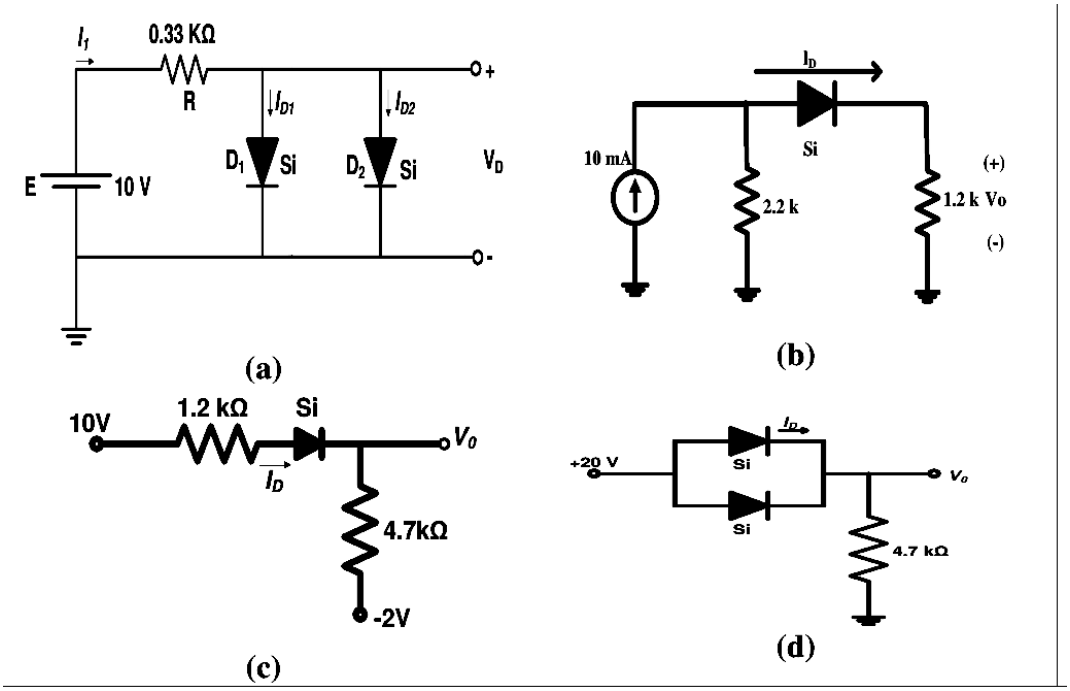
(b) Reversed Ge is OFF; Si is ON

Current flows from 20 V down through Si and 2.2 kΩ to the 4 V terminal:

$$V_o = 20 - 0.7 = 19.3\ \text{V}, \quad I_D = \frac{19.3 - 4}{2.2\ \text{k}\Omega} = 6.95\ \text{mA}$$

(a) $V_o = 11.3$ V, $I_D = 2.40$ mA (b) $V_o = 19.3$ V, $I_D = 6.95$ mA

Check: in Boylestad's original figure the bottom terminal is −4 V; then $I_D = (19.3 + 4)/2.2\ \text{k} = 10.59$ mA. Read the sign on your paper.



QB A-L6 circuits (a) to (d)

Solved

QB A-L6

Four circuits (a)-(d): find the marked currents and voltages.

(a) 10 V, 0.33 kΩ, two Si diodes in parallel

Both ON, identical, so they share the current equally:

$$V_o = 0.7\ \text{V},\ I_1 = \frac{10 - 0.7}{0.33\ \text{k}\Omega} = 28.18\ \text{mA},\ I_{D1} = I_{D2} = 14.09\ \text{mA}$$

(b) 10 mA source, 2.2 kΩ in parallel with (Si + 1.2 kΩ)

Voltage across 2.2 kΩ = 0.7 + 1.2k·I_D, and 10 mA = (that voltage)/2.2 k + I_D:

$$I_D = \frac{10\ \text{mA} \times 2.2\ \text{k}\Omega - 0.7}{2.2\ \text{k}\Omega + 1.2\ \text{k}\Omega} = 6.26\ \text{mA},\ V_o = 1.2\ \text{k}\Omega \times I_D = 7.52\ \text{V}$$

(c) 10 V, 1.2 kΩ, Si, 4.7 kΩ to −2 V

Total drive = 10 − (−2) = 12 V:

$$I_D = \frac{10 - 0.7 + 2}{1.2 + 4.7\ \text{k}\Omega} = 1.915\ \text{mA},\ V_o = -2 + 4.7\ \text{k}\Omega \times I_D = 7.00\ \text{V}$$

(d) 20 V, two Si in parallel, 4.7 kΩ

$$V_o = 20 - 0.7 = 19.3\ \text{V},\ I = 4.11\ \text{mA},\ I_D = \tfrac{1}{2} I = 2.05\ \text{mA}$$

(a) 14.09 mA each · (b) V_o = 7.52 V · (c) V_o = 7.00 V · (d) I_D = 2.05 mA

REMEMBER

1. Si 0.7 V, Ge 0.3 V, GaAs 1.2 V.
2. A diode whose arrow points against the current is OFF (open circuit).
3. Series diodes: any one OFF ⇒ no current anywhere in that branch.
4. Equal parallel diodes share current equally; unequal ones: lowest drop wins.

IF THE QUESTION...

- **If** the supply is smaller than the diode drop (0.5 V on Si) → diode OFF, I = 0, all the supply appears across the diode
- **If** a resistor goes to a negative supply → drive voltage = top − bottom (e.g. 10 − (−2) = 12 V)
- **If** a current source feeds parallel branches → write KCL at the top node (A-L6 b)

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

2.1-1 APPLY 12 V, Ge diode, 1 kΩ in series. Find I.

2.1-2 NEW SETTING A 5 V supply feeds an LED (2 V) and a Si diode in series, then 150 Ω. Find I.

ANSWERS — COVER UNTIL YOU HAVE TRIED

APPLY **2.1-1** (12 − 0.3)/1 k = 11.7 mA.

NEW SETTING **2.1-2** (5 − 2 − 0.7)/150 = 15.3 mA.

3 Rectifiers: half-wave, centre-tap and bridge

UNDERSTAND *how one or two diodes turn AC into one-way DC* · **RECALL EXACTLY** *V_{dc}, I_{rms}, ripple factor, efficiency, PIV and TUF for each* · **BE ABLE TO** *derive them and solve the transformer sums.*

3.1 Half-wave rectifier (HWR)

QB: N-L8, M-S12, M-S14

▣ **DEFINITION**

A **rectifier** converts AC into pulsating (unidirectional) DC. A **half-wave rectifier** uses one diode and passes only one half of each AC cycle to the load.

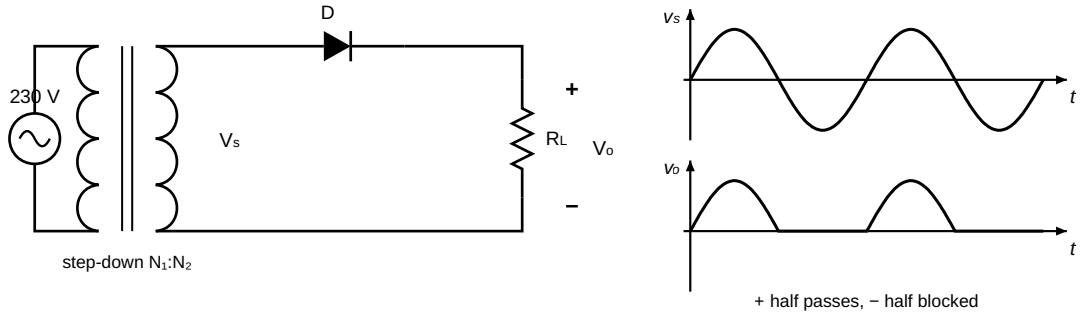


Fig 3.1 — Half-wave rectifier and its waveforms

1. **Circuit:** step-down transformer, one diode D in series with the load R_L.
2. **Positive half-cycle:** top of the secondary is +, D is forward biased (ON), current flows through R_L: $v_o = v_s$.
3. **Negative half-cycle:** D is reverse biased (OFF), no current: $v_o = 0$. The diode must withstand V_m (PIV = V_m).
4. **Result:** one pulse per cycle, output frequency = input frequency (50 Hz).

Applications (M-S12): cheap low-current supplies, battery chargers for small cells, signal detection (AM demodulator), simple power adapters. **Disadvantages (M-S14):** high ripple (121%), low efficiency (40.6%), poor TUF (0.287), DC saturation of the transformer core, needs a big filter.

3.2 Deriving the HWR formulas

QB: M-L16, A-L13, N-L9, M-L10

The standard derivation question: average (DC) value, RMS value, ripple factor, efficiency, PIV.

You'll see it as: "derive average dc output voltage, dc current, ripple factor, efficiency" · "include derivations for average and RMS current"

THE IDEA

1. Write the output: i = I_m sinθ for 0 < θ < π, and 0 for π < θ < 2π.
2. Average = area under one cycle ÷ 2π.
3. RMS = square root of the mean of i^2 over one cycle.
4. Ripple factor and efficiency follow from I_{dc} and I_{rms}.

THE FORMULA

$$I_{dc} = \frac{1}{2\pi}\int_0^{\pi} I_m \sin\theta \, d\theta = \frac{I_m}{\pi}, \qquad I_m = \frac{V_m}{r_f + R_L}$$

Golden rule: average over the WHOLE cycle (divide by 2π), even though current flows for only half of it.

Solved example

Derivation

Derive V_{dc}, I_{rms}, ripple factor γ, efficiency η and PIV for the HWR.

Step 1: average (DC) current and voltage

$$I_{dc} = \frac{I_m}{2\pi}\Big[-\cos\theta\Big]_0^{\pi} = \frac{I_m}{2\pi}(1+1) = \frac{I_m}{\pi}$$

Multiply by R_L (ideal diode, $r_f = 0$):

$$V_{dc} = I_{dc}R_L = \frac{V_m}{\pi} = 0.318 V_m$$

Step 2: RMS current

$$I_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} I_m^2 \sin^2\theta \, d\theta} = \sqrt{\frac{I_m^2}{2\pi}\cdot\frac{\pi}{2}} = \frac{I_m}{2}$$

Step 3: ripple factor

Ripple factor = RMS of AC part ÷ DC part:

$$\gamma = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} = 1.21$$

Step 4: efficiency

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (r_f + R_L)} = \frac{(I_m/\pi)^2 R_L}{(I_m/2)^2 (r_f + R_L)} = \frac{4}{\pi^2}\cdot\frac{R_L}{r_f + R_L}$$

Maximum when $r_f \ll R_L$:

$$\eta_{max} = \frac{4}{\pi^2} = 0.405 = 40.6\%$$

Step 5: PIV

In the negative half-cycle the diode is OFF and the whole secondary peak appears across it:

$$\text{PIV} = V_m$$

$V_{dc} = V_m/\pi$ · $I_{rms} = I_m/2$ · γ = 1.21 (121%) · η = 40.6% · PIV = V_m

Check: $V_{rms} = V_m/2$ for HWR, $V_m/\sqrt{2}$ for full-wave: half the cycle is missing, so the RMS is smaller.

HOW TO WRITE IT IN THE EXAM

Write the waveform equation, the integral, the result line, and box each formula. Draw the circuit and waveforms first: they carry marks.

3.3 Rectifier figures of merit (definitions)

QB: M-S11, A-S3, M-S13, N-S13, M-S15, N-S11, A-S12, N-S12, A-S13, A-S15, A-S16, N-S15

▣ **DEFINITIONS**

PIV (peak inverse voltage): the maximum reverse voltage that appears across a diode when it is not conducting. The diode's rating must exceed it. HWR: V_m; centre-tap: $2V_m$; bridge: V_m.

Ripple: the AC component left in the rectified output. **Ripple factor** γ = (RMS value of AC component) / (DC value).

Rectification efficiency η = DC power to load / AC power input.

TUF (transformer utilisation factor) = DC power to load / AC (VA) rating of the transformer secondary.

Voltage regulation = $(V_{no\text{-}load} - V_{full\text{-}load}) / V_{full\text{-}load} \times 100\%$; ideal value 0%.

★ **KEY POINT**

Load regulation vs diode resistance (A-S13): for the HWR, %regulation = $(r_f/R_L) \times 100$. The larger the diode's resistance, the more the output falls under load, so regulation gets **worse**.

№ **HIGH-YIELD NUMBER**

Ripple: HWR **1.21 (121%)**, full-wave **0.482 (48.2%)**. Efficiency: HWR **40.6%**, full-wave **81.2%**. TUF: HWR 0.287, centre-tap 0.693, bridge 0.812.

3.4 Centre-tap full-wave rectifier

QB: M-L14, A-L14, N-L18

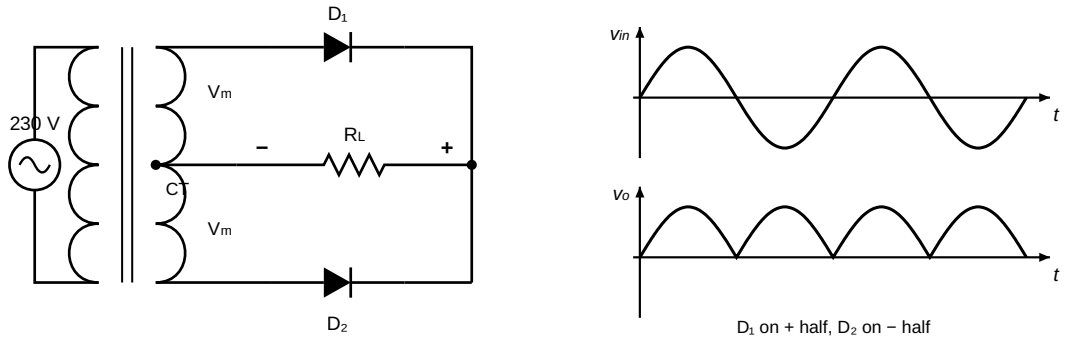


Fig 3.2 — Centre-tap full-wave rectifier (CT = centre tap)

1. **Circuit:** centre-tapped transformer, two diodes D_1, D_2, load between the joined cathodes and the centre tap.

2. **Positive half:** top end +, D_1 ON, D_2 OFF; current through R_L from + to −.

3. **Negative half:** bottom end +, D_2 ON, D_1 OFF; current through R_L in the **same** direction.
4. **PIV = $2V_m$:** the OFF diode sees the full secondary (V_m + V_m).
5. Output: two pulses per cycle, frequency 2f = 100 Hz.

Solved example

Derivation

Efficiency and ripple factor of the full-wave rectifier (same for centre-tap and bridge).

Average and RMS

Two half-sines per cycle, so the area doubles:

$$I_{dc} = \frac{2I_m}{\pi}, \quad V_{dc} = \frac{2V_m}{\pi} = 0.636V_m, \quad I_{rms} = \frac{I_m}{\sqrt{2}}$$

Ripple factor

$$\gamma = \sqrt{\left(\frac{I_m/\sqrt{2}}{2I_m/\pi}\right)^2 - 1} = \sqrt{\frac{\pi^2}{8} - 1} = 0.483$$

Efficiency

$$\eta = \frac{(2I_m/\pi)^2 R_L}{(I_m/\sqrt{2})^2 (r_f + R_L)} = \frac{8}{\pi^2} \cdot \frac{R_L}{r_f + R_L}, \quad \eta_{max} = \frac{8}{\pi^2} = 81.2\%$$

Twice the HWR (A-L16)

$$\frac{\eta_{FWR}}{\eta_{HWR}} = \frac{8/\pi^2}{4/\pi^2} = 2$$

> $V_{dc} = 2V_m/\pi$ · $\gamma = 0.482$ · $\eta_{max} = 81.2\%$ (= 2 × 40.6%)

3.5 Full-wave bridge rectifier

QB: M-L13, A-L10, N-L15, N-L10, A-L16, A-S17, A-L20

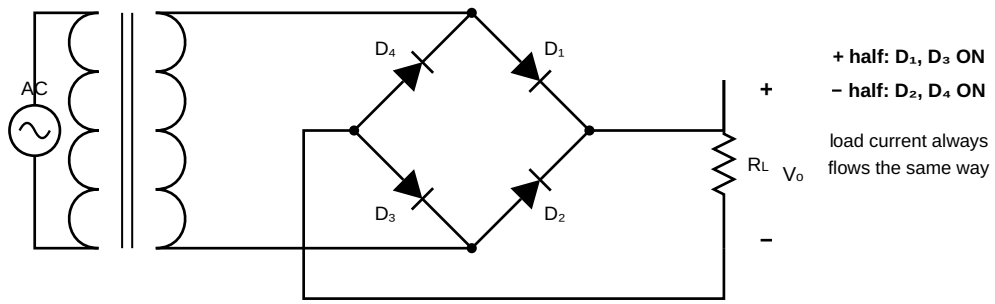


Fig 3.3 — Full-wave bridge rectifier

1. **Circuit:** four diodes in a bridge; AC to one diagonal, load across the other. No centre tap needed.
2. **Positive half:** D_1 and D_3 forward biased, current flows $D_1 \rightarrow R_L \rightarrow D_3$.

3. **Negative half:** D_2 and D_4 conduct, current through R_L in the same direction.
4. Two diodes conduct at a time, so a real bridge loses 2 × 0.7 = 1.4 V.
5. **PIV = V_m** per diode. $V_{dc} = 2V_m/\pi$, $V_{rms} = V_m/\sqrt{2}$, γ = 0.482, η = 81.2%.

> ★ KEY POINT
>
> **Bridge over centre-tap (A-S17):** no centre-tapped transformer needed (smaller, cheaper); PIV only V_m; higher TUF (0.812 vs 0.693). Cost: four diodes and a 1.4 V drop.

3.6 Comparing the three rectifiers

QB: M-L18, N-S16, M-S10, A-S11, N-L15, M-L10

Table 3.1 — Comparison (M-L18)

Parameter	Half-wave	Centre-tap	Bridge
Diodes	1	2	4
Transformer	ordinary	centre-tapped	ordinary (or none)
V_{dc}	$V_m/\pi = 0.318V_m$	$2V_m/\pi = 0.636V_m$	$2V_m/\pi$
I_{rms}	$I_m/2$	$I_m/\sqrt{2}$	$I_m/\sqrt{2}$
Ripple factor	1.21	0.482	0.482
Max efficiency	40.6%	81.2%	81.2%
PIV	V_m	$2V_m$	V_m
Output frequency	f	2f	2f
TUF	0.287	0.693	0.812

HWR vs full-wave in one line (M-S10, A-S11): HWR uses one half of each cycle (one diode, 40.6%, ripple 1.21); full-wave uses both halves (two or four diodes, 81.2%, ripple 0.482). **Two advantages of full-wave (N-S15):** double the efficiency and DC output; much lower ripple, so filtering is easier.

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

3.6-1 RECALL PIV of each rectifier?

3.6-2 TELL APART Why is the centre-tap PIV $2V_m$ but the bridge only V_m?

3.6-3 APPLY 50 Hz mains into a bridge rectifier. Ripple frequency?

3.7 Rectifier sums with a transformer

QB: M-L11, A-L11, N-L17, M-L15, N-L12, N-L13, A-L15, M-L17, A-L17, N-L14

Mains through a step-down transformer into a rectifier; find V_{dc}, PIV, currents, powers, efficiency.

You'll see it as: "230 V through a transformer of turns ratio 10 : 1" · "diode resistance r_f = 20 Ω" · "find PIV / output frequency"

THE IDEA

1. Secondary RMS = primary RMS ÷ turns ratio (230 V is an RMS value).
2. Peak $V_m = \sqrt{2} \times V_{rms}$.
3. $I_m = V_m/(r_f + R_L)$; then use the HWR or FWR formula for I_{dc}, I_{rms}.
4. $P_{dc} = I_{dc}^2 R_L$, $P_{ac} = I_{rms}^2(r_f + R_L)$, $\eta = P_{dc}/P_{ac}$.

THE FORMULA

$$V_s = \frac{V_p}{N_1/N_2}, \quad V_m = \sqrt{2}\,V_s, \quad V_{dc} = \frac{V_m}{\pi}\ (\text{HWR}),\ \frac{2V_m}{\pi}\ (\text{FWR})$$

Golden rule: turn the 230 V RMS into a PEAK before using any rectifier formula.

Solved example

QB M-L11 / A-L11 / N-L17

230 V AC through a 10 : 1 transformer into an ideal HWR. Find V_{dc} and PIV.

Step 1: secondary

$$V_s = \frac{230}{10} = 23\ \text{V (rms)}, \quad V_m = \sqrt{2} \times 23 = 32.53\ \text{V}$$

Step 2: DC output

$$V_{dc} = \frac{V_m}{\pi} = \frac{32.53}{\pi} = 10.35\ \text{V}$$

Step 3: PIV

$$\text{PIV} = V_m = 32.53\ \text{V}$$

V_{dc} = 10.35 V, PIV = 32.53 V

Check: V_{dc} is about a third of V_m, as expected for an HWR.

HOW TO WRITE IT IN THE EXAM

Quantity	*Formula*	*Value*
V_s	230/10	23 V
V_m	$\sqrt{2}\ V_s$	32.53 V
V_{dc}	V_m/π	10.35 V

Quantity	Formula	Value
PIV	V_m	32.53 V

Hence V_{dc} = 10.35 V and PIV = 32.53 V.

More from the question bank, solved

Solved

QB M-L15

r_f = 20 Ω, v = 50 sin ωt, R_L = 800 Ω, HWR. Find I_m, I_{dc}, I_{rms}, P_{ac}, P_{dc}, V_{dc}, η.

Working

$$I_m = \frac{50}{20+800} = 60.98\ \text{mA},\ I_{dc} = \frac{I_m}{\pi} = 19.41\ \text{mA},\ I_{rms} = \frac{I_m}{2} = 30.49\ \text{mA}$$
$$P_{ac} = I_{rms}^2(r_f + R_L) = (0.03049)^2 \times 820 = 0.762\ \text{W}$$
$$P_{dc} = I_{dc}^2 R_L = (0.01941)^2 \times 800 = 0.301\ \text{W}$$
$$V_{dc} = I_{dc}R_L = 15.53\ \text{V}, \quad \eta = \frac{0.301}{0.762} = 39.5\%$$

I_m = 60.98 mA, I_{dc} = 19.41 mA, I_{rms} = 30.49 mA, P_{ac} = 0.762 W, P_{dc} = 0.301 W, V_{dc} = 15.53 V, η = 39.5%

Check: η must come out just below 40.6% because r_f is small: 40.6 × 800/820 = 39.6%.

Solved

QB N-L12

HWR must give 50 V DC across 800 Ω; diode resistance 25 Ω. Find the AC voltage needed.

Working

$$I_{dc} = \frac{50}{800} = 62.5\ \text{mA}, \quad I_m = \pi I_{dc} = 0.1963\ \text{A}$$
$$V_m = I_m(r_f + R_L) = 0.1963 \times 825 = 162.0\ \text{V}, \quad V_{rms} = \frac{V_m}{\sqrt{2}} = 114.5\ \text{V}$$

AC input needed ≈ 114.5 V (rms), peak 162.0 V

Solved

QB N-L13

AC input to an HWR is 100 W; DC output is 40 W. (a) Efficiency? (b) What happens to the other 60 W?

Working

$$\eta = \frac{P_{dc}}{P_{ac}} = \frac{40}{100} = 40\%$$

(b) It is NOT all lost as heat. The 100 W is shared equally between the two half-cycles. The 50 W of the negative halves is never drawn (the diode is OFF). Of the 50 W in the positive halves, 40 W becomes DC; the rest is the AC ripple power in the load plus a little heat in the diode.

(a) 40% (b) 50 W is never taken from the supply; only about 10 W goes into ripple and diode heat

Solved

QB A-L15

230 V, 60 Hz, 5 : 1 centre-tap transformer, full-wave, R_L = 900 Ω, diode + winding resistance 100 Ω. Find V_{dc}, I_{dc}, P_{dc}, PIV.

Working

Whole secondary = 230/5 = 46 V; each half = 23 V rms:

$$V_m = \sqrt{2} \times 23 = 32.53 \text{ V}, \quad I_m = \frac{32.53}{900 + 100} = 32.53 \text{ mA}$$

$$I_{dc} = \frac{2I_m}{\pi} = 20.71 \text{ mA}, \; V_{dc} = I_{dc}R_L = 18.64 \text{ V}, \; P_{dc} = I_{dc}^2 R_L = 0.386 \text{ W}$$

Centre-tap:

$$\text{PIV} = 2V_m = 65.1 \text{ V}$$

V_{dc} = 18.64 V, I_{dc} = 20.71 mA, P_{dc} = 0.386 W, PIV = 65.1 V

Solved

QB M-L17 / A-L17

Bridge, 120 V rms input, R_L = 1 kΩ, Si diodes. (a) V_{dc} (b) PIV (c) max diode current (d) diode power rating.

Working

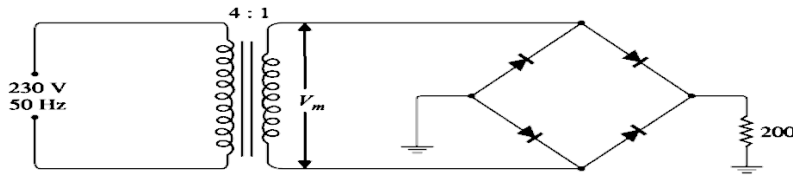
Two diodes conduct, so subtract 1.4 V from the peak:

$$V_m = \sqrt{2} \times 120 = 169.71 \text{ V}, \; V_{L(peak)} = 169.71 - 1.4 = 168.31 \text{ V}$$

$$V_{dc} = \frac{2 \times 168.31}{\pi} = 107.1 \text{ V}, \quad \text{PIV} \geq V_m - 0.7 = 169.0 \text{ V}$$

$$I_{D(max)} = \frac{168.31}{1 \text{ k}\Omega} = 168.3 \text{ mA}, \quad P_D = 0.7 \times 0.1683 = 117.8 \text{ mW}$$

V_{dc} = 107.1 V, PIV ≥ 169 V, $I_{D(max)}$ = 168.3 mA, P_D = 117.8 mW



QB N-L14 circuit

Solved

QB N-L14

230 V, 50 Hz, 4 : 1 transformer, ideal bridge, 200 Ω. Find V_{dc}, PIV, output frequency.

Working

$$V_s = \frac{230}{4} = 57.5 \text{ V}, \; V_m = \sqrt{2} \times 57.5 = 81.32 \text{ V}, \; V_{dc} = \frac{2V_m}{\pi} = 51.77 \text{ V}$$

$$\text{PIV} = V_m = 81.32 \text{ V}, \quad f_{out} = 2 \times 50 = 100 \text{ Hz}$$

V_{dc} = 51.77 V, PIV = 81.32 V, f_{out} = 100 Hz

REMEMBER

1. 230 V mains is RMS. Peak = √2 × RMS.
2. Centre-tap: each HALF of the secondary gives V_m; PIV = $2V_m$.
3. Real bridge: subtract 1.4 V (two diodes); real HWR or centre-tap: subtract 0.7 V.
4. Full-wave output frequency = 2 × supply frequency.

IF THE QUESTION...

- **If** gives a diode resistance r_f → $I_m = V_m/(r_f + R_L)$ and P_{ac} uses $(r_f + R_L)$
- **If** gives the DC output and asks for the AC input → work backwards: I_{dc} → I_m → V_m → V_{rms} (N-L12)
- **If** says “turns ratio 4” → primary : secondary = 4 : 1, so $V_s = V_p/4$

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

3.7-1 APPLY 230 V, 10 : 1, ideal bridge. V_{dc}?

3.7-2 NEW SETTING A phone charger’s transformer gives 9 V rms into a Si bridge. Peak voltage at the load?

3.8 Sketching outputs of unusual bridge circuits

QB: A-L18

A sine of ±100 V feeds a bridge drawn in an unusual way; sketch v_o and find PIV.

You’ll see it as: “sketch the output voltage and current waveforms and find the PIV”

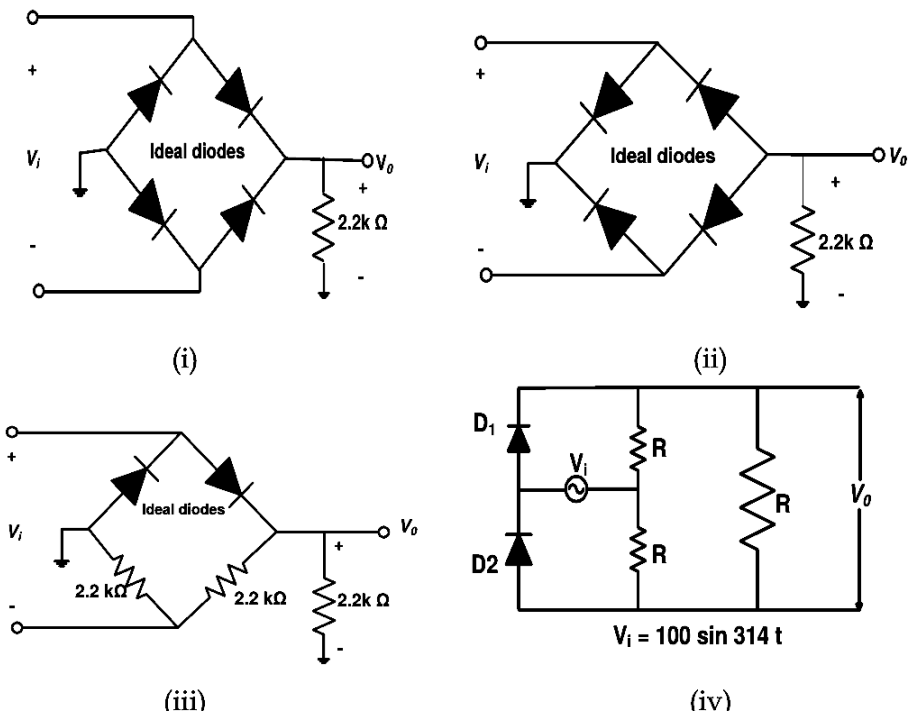
THE IDEA

1. Take the positive half: find which diodes are forward biased and trace the current path through the load.
2. Repeat for the negative half.
3. If resistors replace two diodes, work out the voltage divider.
4. PIV = the largest reverse voltage across an OFF diode.

THE FORMULA

$$V_{dc} = 0.636\, V_{o(peak)}$$

Golden rule: trace the current; the load sees the same polarity in both halves = full-wave.



QB A-L18 circuits (i) to (iv), V_i = 100 V peak, ideal diodes

Solved example

QB A-L18

Sketch v_o and find the PIV for (i)-(iv).

(i) Normal bridge, cathodes meet at V_o

Both halves drive the load from V_o to ground: positive full-wave output, peak 100 V.

$$V_{dc} = 0.636 \times 100 = 63.7 \text{ V},\ \text{PIV} = 100 \text{ V},\ I_{max} = \frac{100}{2.2 \text{ k}\Omega} = 45.5 \text{ mA}$$

(ii) All four diodes reversed

Anodes now meet at V_o, so the load current flows up from ground: NEGATIVE full-wave output, peak −100 V; PIV = 100 V.

(iii) Two diodes replaced by 2.2 kΩ resistors

Each half: one diode ON; the input divides between the load and one 2.2 kΩ in series, so the load gets half:

$$V_{o(peak)} = \frac{100}{2} = 50 \text{ V}, \quad V_{dc} = 0.636 \times 50 = 31.8 \text{ V}, \quad \text{PIV} = 50 \text{ V}$$

(iv) Two diodes and three equal R

Positive half: D_1 ON, the load and one R share V_i equally. Negative half: D_2 ON, same again in the same direction:

$$V_{o(peak)} = 50 \text{ V (full-wave)}, \quad \text{PIV} = 50 \text{ V}, \quad f_{out} = 2 \times \frac{314}{2\pi} = 100 \text{ Hz}$$

(i) +100 V full-wave, PIV 100 V · (ii) −100 V full-wave, PIV 100 V · (iii) 50 V full-wave, PIV 50 V · (iv) 50 V full-wave, PIV 50 V

Check: sketch each output under the input sine: same period, both humps on the same side.

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

3.8-1 APPLY In (i), what is the peak current in each diode?

ANSWERS — COVER UNTIL YOU HAVE TRIED

RECALL **3.6-1** HWR V_m; centre-tap $2V_m$; bridge V_m.

TELL APART **3.6-2** In the centre-tap the OFF diode sees both halves of the secondary in series ($V_m + V_m$); in the bridge each OFF diode is across only one secondary of peak V_m.

APPLY **3.6-3** 100 Hz.

APPLY **3.7-1** $V_m = 32.53$ V, $V_{dc} = 2 \times 32.53/\pi = 20.71$ V.

NEW SETTING **3.7-2** $\sqrt{2} \times 9 - 1.4 = 11.33$ V.

APPLY **3.8-1** 100 V / 2.2 kΩ = 45.5 mA.

4 Filters and the regulated DC power supply

UNDERSTAND *how a capacitor smooths pulsating DC* · **RECALL EXACTLY** *the ripple formula and the four blocks of a supply* · **BE ABLE TO** *calculate ripple and draw the block diagram with waveforms.*

4.1 Filters and the capacitor filter

QB: M-L19, A-L19, A-L20, N-L11, A-S14, M-S25

▣ **DEFINITION**

A **filter** is a circuit that removes or reduces the AC components (ripple) from the pulsating DC output of a rectifier, giving a smoother DC voltage.

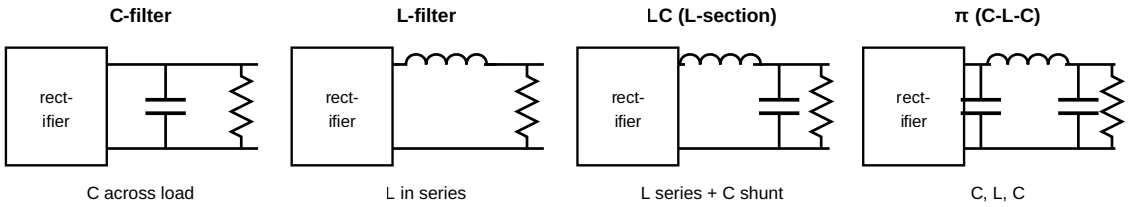


Fig 4.2 — The four filter types (last resistor = load RL)

Elements used (A-S14): capacitor connected in **parallel** (shunt) with the load; **inductor** (choke) connected in **series** with the load. A capacitor passes AC to ground and blocks DC; an inductor passes DC and opposes changes in current.

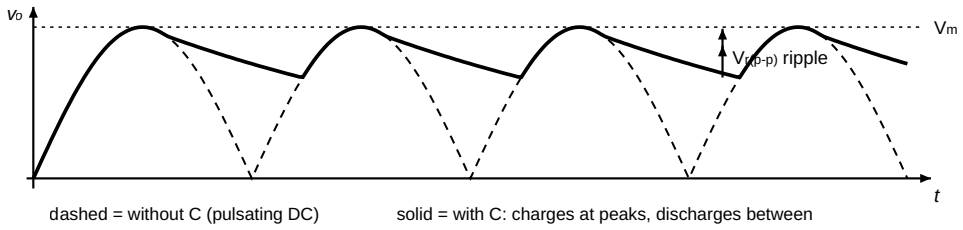


Fig 4.1 — Full-wave output with and without a filter capacitor

1. **Charging:** as the rectified voltage rises, the diode conducts and C charges quickly to almost V_m.
2. **Discharging:** after the peak, the rectified voltage falls below the capacitor voltage, the diodes switch OFF, and C discharges slowly through R_L, supplying the load.
3. **Next peak:** the diodes switch on again and top C up. The output never falls to zero: only a small ripple remains.
4. **Why it reduces ripple:** $X_C = 1/(2\pi fC)$ is small for ripple frequency, so the AC is bypassed; for DC (f = 0) C is an open circuit, so DC goes to the load.

★ KEY POINT

Why a filter capacitor is used after a rectifier (M-S25): rectifier output is pulsating DC with large ripple; the capacitor stores charge at the peaks and releases it between them, giving nearly constant DC. A larger C or larger R_L gives less ripple.

4.2 Ripple with a capacitor filter

QB: M-L12, A-L12, N-L16

Bridge rectifier plus filter capacitor: find peak output, DC output and ripple voltage.

You'll see it as: "transformer 10 : 1, filter capacitor 470 μF; ripple voltage for a load of 1 kΩ"

THE IDEA

1. Find V_m from the secondary RMS.
2. Ripple (peak-to-peak) = $V_m/(2fCR_L)$ for full-wave.
3. DC output = $V_m - V_{r(p\text{-}p)}/2$.

THE FORMULA

$$V_{r(p\text{-}p)} = \frac{I_{dc}}{2fC} \approx \frac{V_m}{2fCR_L}\ (\text{full-wave}), \qquad V_{dc} = V_m - \frac{V_{r(p\text{-}p)}}{2}$$

Golden rule: full-wave ripple uses 2f (100 Hz); half-wave uses f.

Solved example

QB M-L12 / A-L12 / N-L16

Bridge, 220 V, 50 Hz, 10 : 1 transformer, C = 470 μF, R_L = 1 kΩ. Find peak output, DC output, ripple.

Step 1: peak

$$V_s = \frac{220}{10} = 22\ \text{V}, \quad V_m = \sqrt{2} \times 22 = 31.11\ \text{V}$$

Step 2: ripple

$$V_{r(p\text{-}p)} = \frac{31.11}{2 \times 50 \times 470 \times 10^{-6} \times 1000} = \frac{31.11}{47} = 0.662\ \text{V}$$

Step 3: DC output

$$V_{dc} = 31.11 - \frac{0.662}{2} = 30.78\ \text{V}$$

V_m = 31.11 V · V_{dc} ≈ 30.78 V · $V_{r(p\text{-}p)}$ ≈ 0.66 V

Check: without the capacitor V_{dc} would be only $2V_m/\pi$ = 19.81 V. With Si diodes, use V_m = 29.71 V instead.

HOW TO WRITE IT IN THE EXAM

Quantity	*Value*
Peak output	31.11 V

Quantity	*Value*
Ripple (p-p)	0.662 V
DC output	30.78 V
Ripple (rms) = $V_{r(p\text{-}p)}/(2\sqrt{3})$	0.191 V

State “ideal diodes assumed” and box the three answers.

REMEMBER

1. Ripple falls when C, R_L or frequency rise.
2. Full-wave: 2f; half-wave: f.
3. With a good filter, V_{dc} is close to V_m.

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

4.2-1 APPLY Same circuit, but a half-wave rectifier. Ripple?

4.2-2 NEW SETTING Load current doubles (R_L = 500 Ω). What happens to ripple, and why?

4.3 The regulated DC power supply

QB: M-L28, A-L28, M-L30, A-S29, A-S26, A-S27

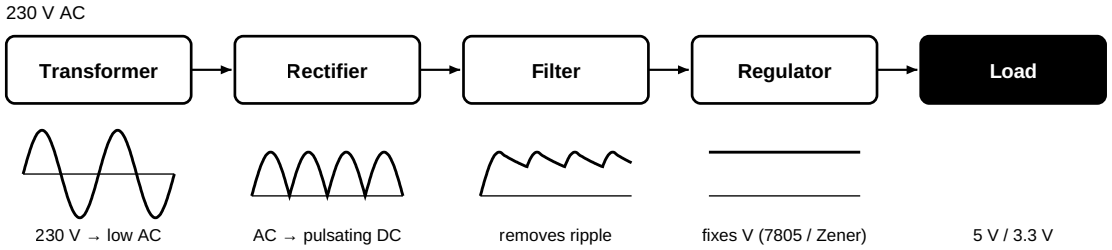


Fig 4.3 — Regulated DC power supply: block diagram with the output waveform of each block

► **Transformer:** steps 230 V AC down to a low AC voltage (and isolates the circuit from the mains).

► **Rectifier:** converts AC to pulsating DC (bridge, usually).

► **Filter:** capacitor removes most of the ripple.

► **Regulator:** holds the output constant (Zener, or IC 7805 for 5 V, AMS1117 for 3.3 V) even if mains or load changes.

▣ **DEFINITIONS**

Voltage regulator: a circuit that keeps the output DC voltage constant despite changes in input voltage or load current (A-S27). **Unregulated** supply: output changes with mains and load. **Regulated** supply: output stays fixed (A-S29).

Why DC is needed (A-S26): transistors, ICs and microcontrollers need a fixed polarity and constant voltage to bias and switch correctly; AC would reverse them 50 times a second.

ANSWERS — COVER UNTIL YOU HAVE TRIED

APPLY **4.2-1** $V_m/(fCR_L) = 31.11/23.5 = 1.32$ V (double).

NEW SETTING **4.2-2** Ripple doubles: C discharges twice as fast between peaks.

5 The Zener diode and the Zener regulator

> **UNDERSTAND** *the two breakdown mechanisms and why the Zener holds its voltage* · **RECALL EXACTLY** *the four regulator cases* · **BE ABLE TO** *solve every Zener network: ON/OFF test, currents, power, input range, series resistor.*

5.1 Breakdown mechanisms: Zener and avalanche

QB: M-L20, A-L21, N-L25, A-L22, A-L23, M-S16, A-S19, N-S17, M-S21

▣ **DEFINITION**

A **Zener (breakdown) diode** is a heavily doped P-N junction diode designed to work safely in reverse breakdown, where it keeps an almost constant voltage V_Z across it as long as its current and power ratings are not exceeded.

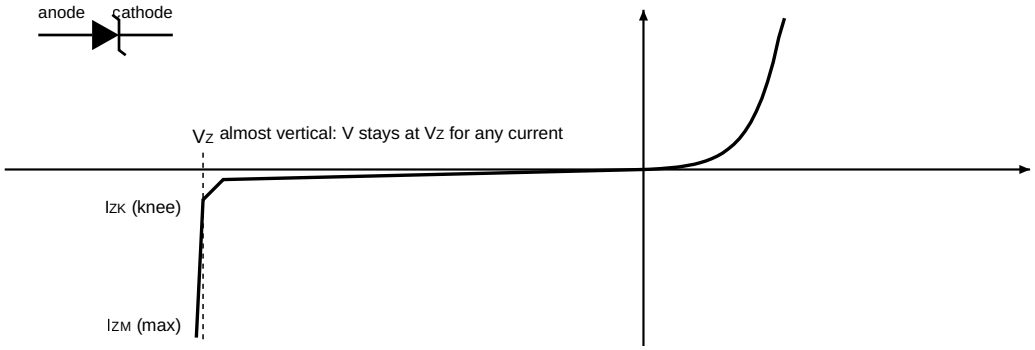


Fig 5.1 — Zener diode symbol and V-I characteristic

Zener breakdown (low V_Z, below about 5 V)

1. The junction is **heavily doped**, so the depletion layer is **very thin**.
2. A small reverse voltage across a thin layer gives a **very strong electric field** (about 3×10^7 V/m).
3. The field pulls electrons straight out of their covalent bonds: they **tunnel** from the valence band to the conduction band.
4. Many carriers appear at once, so reverse current rises sharply at a fixed voltage.

Avalanche breakdown (high V_Z, above about 6 V)

1. The junction is **lightly doped**, so the depletion layer is **wide**.
2. Minority carriers are accelerated by the field and gain high kinetic energy.
3. They **collide** with atoms and knock out new electron-hole pairs (impact ionisation).

4 The new carriers are accelerated too and collide again: a chain reaction (avalanche) and a sharp rise in current.

	Zener breakdown	Avalanche breakdown
LOOK ALIKE BECAUSE	both give a sudden rise of reverse current at a fixed reverse voltage	
DOPING	Heavy	Light
DEPLETION LAYER	Very thin	Wide
MECHANISM	Quantum tunnelling by strong field	Impact ionisation (collisions)
BREAKDOWN VOLTAGE	Low (< 5 V)	High (> 6 V)
V-I CURVE AT BREAKDOWN	Sharper knee, slightly rounded	Very sharp
TEMPERATURE COEFFICIENT	**Negative**	**Positive**
THE DIFFERENCE THAT MATTERS	**Field pulls electrons out**	**Collisions knock electrons out**

★ KEY POINT

Why the temperature coefficients differ (A-L22, N-S17, M-S21):
Avalanche: positive. Heat makes the lattice vibrate more, so carriers collide more often and gain less energy between collisions. A higher voltage is needed to make them ionise: V_{BR} rises with temperature.
Zener: negative. Heat gives valence electrons extra energy, so they tunnel out at a weaker field: V_Z falls with temperature. Near 5-6 V the two effects cancel (almost zero coefficient).

5.2 Zener ratings and terms

QB: M-S17, M-S18, M-S19, M-S20, A-S21, N-S18, N-S19, A-S22

▣ DEFINITIONS

Zener voltage V_Z (also the answer to M-S19): the reverse voltage at which the Zener breaks down and conducts, and which it then holds almost constant.
Knee Zener current I_{ZK}: the minimum reverse current needed to keep the diode in breakdown (regulating). Below it, regulation is lost.
Maximum Zener current I_{ZM} = $P_{Z(max)}/V_Z$.
Derating factor: the amount (mW per °C) by which the maximum power rating must be reduced for each degree above 25 °C.

Symbol and terminals (N-S18): a diode symbol whose bar has bent ends (Fig 5.1, top left). Triangle side = anode, bar = cathode; in a regulator the cathode goes to the positive side. **At V_Z (N-S19)** the reverse current rises sharply while the voltage stays at V_Z; a series resistor must limit it.

A-S22: a 4.5 V, 1.5 W Zener: $I_{ZM} = P/V_Z = 1.5/4.5 =$ **0.333 A (333 mA)**.

Applications (M-S17): voltage regulators · reference voltages · overvoltage and surge protection · clippers and wave shapers · meter protection.

5.3 The Zener as a shunt voltage regulator

QB: M-L25, A-L27, N-L19, M-L22, A-L24, N-L23, N-S20

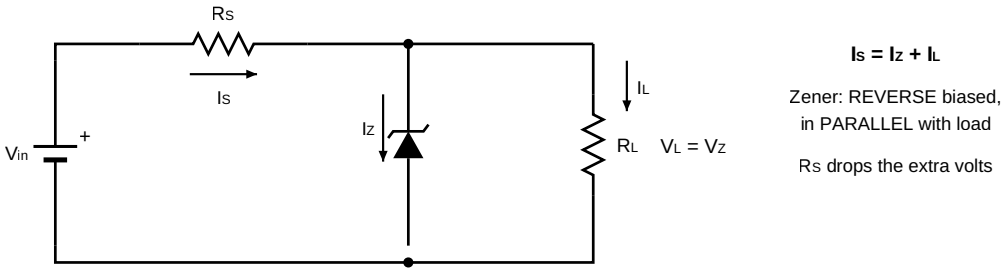


Fig 5.2 — Zener shunt voltage regulator

The Zener is connected in **reverse bias** and in **parallel (shunt)** with the load; a series resistor R_S carries the total current $I_S = I_Z + I_L$. While the Zener is in breakdown, $V_L = V_Z$. Any change is absorbed by I_Z:

Table 5.1 — The four cases (valid while $I_{ZK} < I_Z < I_{ZM}$)

Case	What changes	What the Zener does	V_L
1	V_{in} rises	I_S rises; extra current goes into the Zener (I_Z ↑)	stays V_Z
2	V_{in} falls	I_S falls; I_Z falls to compensate	stays V_Z
3	Load current rises	I_Z falls by the same amount (I_S fixed)	stays V_Z
4	Load current falls	I_Z rises by the same amount	stays V_Z

Why regulation is needed (M-L25, A-L27): mains voltage and load current keep changing; ICs and microcontrollers need a fixed voltage or they reset, misread or get damaged.

	Advantages	Limitations
LOOK ALIKE BECAUSE	N-S20 asks for two of each	
1	Simple: one resistor and one diode	Wastes power in R_S and the Zener (low efficiency)
2	Cheap and small	Only small load currents; output shifts slightly with I_Z
THE DIFFERENCE THAT MATTERS	**Good for references and low-power loads**	**Use an IC regulator (7805) for bigger loads**

5.4 Solving Zener networks

QB: M-L21, M-L22, M-L23, M-L24, A-L24, A-L25, A-L26, N-L21, N-L22, N-L23, N-L24, N-L26

Fixed V_{in} and R_L (is it ON?), varying V_{in} (range of I_Z or V_{in}), or design of R_S.

You'll see it as: "determine V_L, V_R, I_Z, P_Z" · "max and min Zener current" · "range of input voltage for the Zener to remain ON" · "calculate the series resistance R"

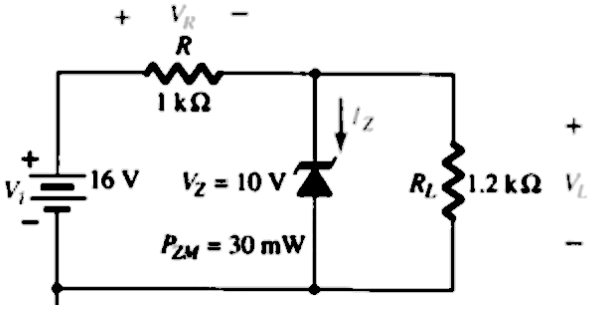
THE IDEA

1. **ON/OFF test:** remove the Zener; find the voltage V across R_L by the voltage divider.
2. If $V \geq V_Z$: Zener ON, $V_L = V_Z$. If $V < V_Z$: Zener OFF, $I_Z = 0$, $V_L = V$.
3. When ON: $I_R = (V_{in} - V_Z)/R$, $I_L = V_Z/R_L$, $I_Z = I_R - I_L$, $P_Z = V_Z I_Z$.
4. Check $P_Z \leq P_{ZM}$.

THE FORMULA

$$V = \frac{R_L}{R + R_L} V_{in} \ \text{(test)}, \qquad I_Z = I_R - I_L = \frac{V_{in} - V_Z}{R} - \frac{V_Z}{R_L}$$

Golden rule: always do the ON/OFF test first: the Zener is not ON just because it is drawn.



QB M-L21 / N-L22

Solved example

QB M-L21 / N-L22

V_i = 16 V, R = 1 kΩ, V_Z = 10 V, P_{ZM} = 30 mW. (a) R_L = 1.2 kΩ (b) R_L = 3 kΩ. Find V_L, V_R, I_Z, P_Z.

(a) Step 1: ON/OFF test

Remove the Zener:

$$V = \frac{1.2}{1 + 1.2} \times 16 = 8.73 \text{ V} < 10 \text{ V}$$

So the Zener is OFF (open):

$$V_L = 8.73 \text{ V},\ V_R = 16 - 8.73 = 7.27 \text{ V},\ I_Z = 0,\ P_Z = 0$$

(b) Step 1: ON/OFF test

$$V = \frac{3}{1 + 3} \times 16 = 12.0 \text{ V} > 10 \text{ V} \Rightarrow \text{ON},\ V_L = 10 \text{ V}$$

(b) Step 2: currents and power

$$V_R = 16 - 10 = 6 \text{ V},\ I_R = \frac{6}{1 \text{ k}\Omega} = 6 \text{ mA},\ I_L = \frac{10}{3 \text{ k}\Omega} = 3.33 \text{ mA}$$

$$I_Z = 6 - 3.33 = 2.67\ \text{mA},\ P_Z = 10 \times 2.67\ \text{mA} = 26.67\ \text{mW}$$

(a) V_L = 8.73 V, V_R = 7.27 V, I_Z = 0, P_Z = 0 (b) V_L = 10 V, V_R = 6 V, I_Z = 2.67 mA, P_Z = 26.67 mW

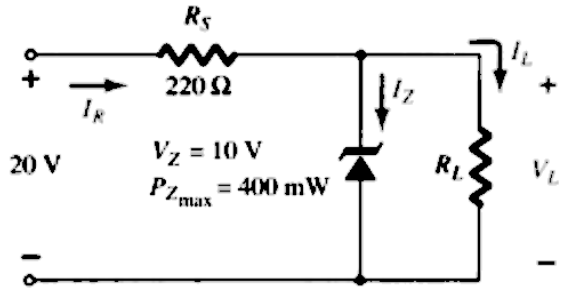
Check: in (b) P_Z = 26.7 mW < 30 mW, so the diode is safe.

HOW TO WRITE IT IN THE EXAM

	Test V	State	V_L	I_Z	P_Z
(a)	8.73 V	OFF	8.73 V	0	0
(b)	12 V	ON	10 V	2.67 mA	26.67 mW

Hence the Zener regulates only when R_L is large enough.

More from the question bank, solved



QB M-L22 / A-L24 / N-L23

Solved

QB M-L22 / A-L24 / N-L23

20 V, R_S = 220 Ω, V_Z = 10 V, P_{Zmax} = 400 mW, R_L = 180 Ω. Find V_L, I_L, I_Z, I_R.

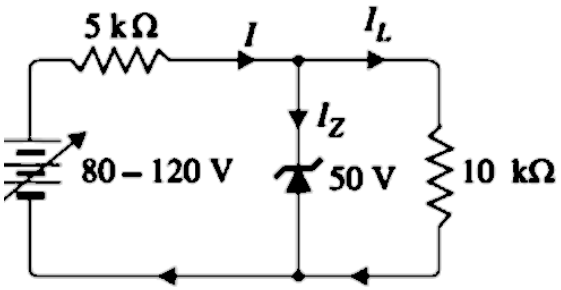
Working

ON/OFF test:

$$V = \frac{180}{220 + 180} \times 20 = 9.0\ \text{V} < 10\ \text{V} \Rightarrow \text{Zener OFF}$$

$$V_L = 9.0\ \text{V},\ I_L = I_R = \frac{20}{400} = 50\ \text{mA},\ I_Z = 0$$

V_L = 9 V, I_L = I_R = 50 mA, I_Z = 0 (the load is too heavy for regulation)



QB M-L23 / A-L25 / N-L24

Solved

QB M-L23 / A-L25 / N-L24

Supply 80-120 V, R = 5 kΩ, V_Z = 50 V, R_L = 10 kΩ. Max and min Zener current.

Working

Load current is fixed by V_Z:

$$I_L = \frac{50}{10\ \text{k}\Omega} = 5\ \text{mA}$$

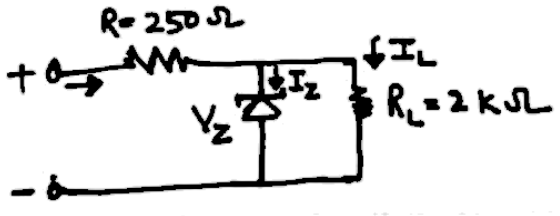
Minimum (80 V):

$$I = \frac{80 - 50}{5\ \text{k}\Omega} = 6\ \text{mA},\ I_{Z(min)} = 6 - 5 = 1\ \text{mA}$$

Maximum (120 V):

$$I = \frac{120 - 50}{5\ \text{k}\Omega} = 14\ \text{mA},\ I_{Z(max)} = 14 - 5 = 9\ \text{mA}$$

$I_{Z(max)}$ = 9 mA, $I_{Z(min)}$ = 1 mA



QB M-L24 / A-L26 / N-L26

Solved

QB M-L24 / A-L26 / N-L26

R = 250 Ω, V_Z = 20 V, I_{Zmax} = 50 mA, R_L = 2 kΩ. Range of V_s for the Zener to stay ON.

Working

$$I_L = \frac{20}{2\ \text{k}\Omega} = 10\ \text{mA}$$

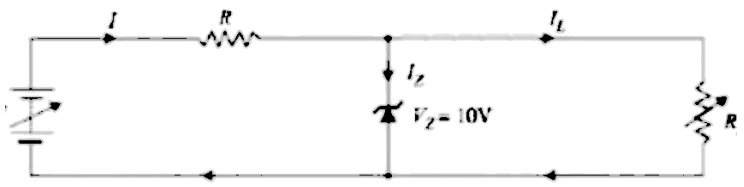
Lowest V_s: Zener just ON, $I_Z \approx 0$ (R_Z = 0, I_{ZK} not given):

$$V_{s(min)} = 20 + 250 \times 10\ \text{mA} = 22.5\ \text{V}$$

Highest V_s: I_Z = 50 mA:

$$V_{s(max)} = 20 + 250 \times (10 + 50)\ \text{mA} = 35.0\ \text{V}$$

22.5 V ≤ V_s ≤ 35 V



QB N-L21

Solved

QB N-L21

V_Z = 10 V; V_{in} 13-16 V; I_L 10-85 mA; $I_{Z(min)}$ = 15 mA. Find R.

Working

Worst case = lowest input AND highest load current; R must still deliver $I_{L(max)} + I_{Z(min)}$:

$$R = \frac{V_{in(min)} - V_Z}{I_{L(max)} + I_{Z(min)}} = \frac{13 - 10}{85 + 15\ \text{mA}} = 30\ \Omega$$

R = 30 Ω

Check: at 16 V and 10 mA load the Zener must carry (16 − 10)/30 − 0.01 = 0.19 A, so choose a Zener rated above 1.9 W.

REMEMBER

1. ON/OFF test first, every time.
2. $I_R = I_Z + I_L$; $V_L = V_Z$ only when ON.
3. $P_Z = V_Z I_Z$, must stay below P_{ZM}.
4. Designing R: use minimum V_{in} with maximum I_L.

IF THE QUESTION...

- **If** asks for the input range → minimum: $I_Z = I_{ZK}$ (or 0); maximum: $I_Z = I_{ZM}$; $V_s = V_Z + R(I_Z + I_L)$
- **If** the supply varies (80-120 V) → I_L is fixed; find I at each end, subtract I_L
- **If** gives P_{ZM} → find $I_{ZM} = P_{ZM}/V_Z$ and check I_Z against it

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

5.4-1 APPLY In M-L22, what is the smallest R_L for which the Zener turns ON?

5.4-2 NEW SETTING Would a 5.1 V Zener with R_S = 100 Ω protect a 3.3 V sensor fed from a 9 V battery?

ANSWERS — COVER UNTIL YOU HAVE TRIED

APPLY **5.4-1** Need $R_L/(220 + R_L) \times 20 = 10$, so R_L = 220 Ω.

NEW SETTING **5.4-2** No: the Zener holds 5.1 V, which is above 3.3 V. Choose a 3.3 V Zener (or a 3.3 V regulator).

6 LEDs and real-world power circuits

> **UNDERSTAND** *how an LED makes light and why it needs a resistor* · **RECALL EXACTLY** $R = (V_S - V_F)/I_F$ · **BE ABLE TO** *explain USB chargers, Arduino/Raspberry Pi supplies, indicators and protection diodes.*

6.1 The light-emitting diode (LED)

QB: M-L26, A-L29, N-L27, M-S22, A-S23, M-S2, A-S24, N-S21, N-S22

▣ **DEFINITION**

An **LED** is a P-N junction diode that emits light when forward biased; it converts electrical energy into light by **electroluminescence**.

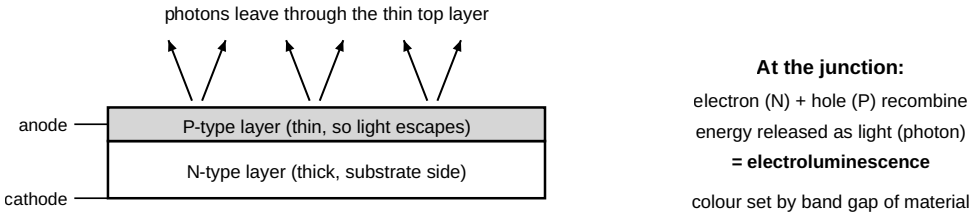


Fig 6.2 — LED construction and working

1. Forward bias: P to +, N to −.
2. Electrons from N and holes from P move to the junction.
3. They **recombine**; each recombination releases energy as a **photon**.
4. The colour (wavelength) is set by the band gap of the material: bigger gap → bluer light, higher V_F.

Construction: a thin P-layer grown over an N-type substrate (so light escapes), made of compound semiconductors such as GaAs (infrared), GaAsP (red/yellow), GaP (green), InGaN (blue/white); mounted under a plastic dome lens; longer lead = anode.

Table 6.1 — LED colours

Colour	Material	Typical V_F
Infrared	GaAs	1.2-1.6 V
Red	GaAsP / AlGaInP	1.8-2.2 V
Green	GaP / InGaN	2.0-3.4 V
Blue / white	InGaN, GaN	3.0-3.6 V

► **Advantages:** low power, long life (tens of thousands of hours), fast switching, small, rugged, little heat, many colours.

► **Applications:** indicator lights, seven-segment displays, traffic lights, LED bulbs and backlights, IR remote controls, optical communication.

► **Precautions (N-S22):** always use a series resistor; connect the right way (reverse breakdown is only about 5 V); never exceed rated current; avoid overheating while soldering; handle with anti-static care.

	Ordinary P-N diode	LED
LOOK ALIKE BECAUSE	both are P-N junctions that conduct in forward bias	
MATERIAL	Si, Ge (indirect band gap)	GaAs, GaAsP, GaN (direct band gap)
FORWARD DROP	0.7 V / 0.3 V	1.2-3.6 V
REVERSE RATING	high (tens to hundreds of V)	low (about 5 V)
USE	rectifying, switching	light, indication
THE DIFFERENCE THAT MATTERS	**Recombination energy → heat**	**Recombination energy → light**

6.2 The current-limiting resistor

QB: M-L27, A-L30, N-L30, M-S23, N-S23, M-S28, A-S20, M-L31

Choosing the series resistor so the LED gets its rated current from a given supply.

You'll see it as: "calculate the series resistor for an LED current of 10 mA" · "why is the resistor necessary"

THE IDEA

1. Subtract the LED drop from the supply: that voltage sits across R.
2. Divide by the wanted current.
3. Pick the next standard value (e.g. 330 Ω) and recheck the current.
4. Power in R = I^2R.

THE FORMULA

$$R = \frac{V_S - V_F}{I_F}, \qquad P_R = I_F^2 R$$

Golden rule: the resistor gets the supply MINUS the LED drop, not the whole supply.

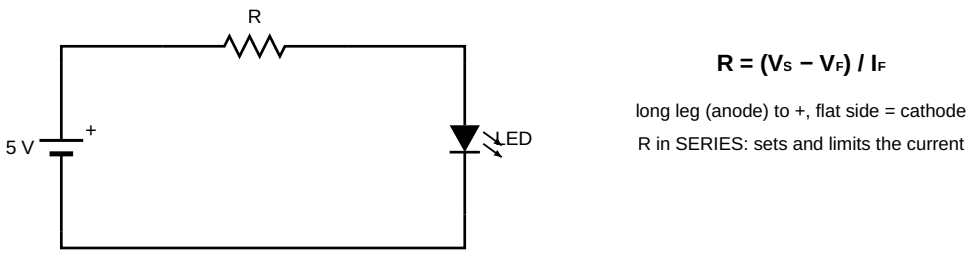


Fig 6.1 — LED with current-limiting resistor

Solved example

QB M-L27 / M-S23 / N-S23

LED V_F = 2 V, supply 5 V, wanted I_F = 10 mA. (a) R? (b) current with 330 Ω? (c) power in R? (d) why R?

(a) Resistor

$$R = \frac{5 - 2}{10 \times 10^{-3}} = 300\ \Omega$$

(b) With the standard 330 Ω

$$I = \frac{5 - 2}{330} = 9.09\ \text{mA}$$

(c) Power

$$P = I^2 R = (0.00909)^2 \times 330 = 27.3\ \text{mW}$$

(d) Why

An LED's forward resistance is very low: past V_F a small rise in voltage makes a huge rise in current that burns it out. R fixes the current.

R = 300 Ω · I = 9.09 mA · P_R = 27.3 mW

Check: 330 Ω is the next standard value above 300 Ω, so the current is slightly below 10 mA: safe.

HOW TO WRITE IT IN THE EXAM

Draw Fig 6.1, write R = (V_S − V_F)/I_F, substitute, box. Add one line on why R is needed.

REMEMBER

1. Without R the LED burns out (M-S28, A-S20).
2. Choose the next standard value ABOVE the calculated one.
3. Indicator LEDs: 2-20 mA.

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

6.2-1 APPLY Blue LED, V_F = 3.2 V, from 5 V at 15 mA. R?

6.2-2 NEW SETTING A 12 V car supply feeds a red LED (2 V) at 20 mA. R and its power?

6.3 USB chargers and Arduino / Raspberry Pi supplies

QB: N-L28, M-S24, A-S25, M-S26, M-L29, M-L30, A-L31, N-L29, M-S27, A-S28

USB charger rectifier

1. 230 V AC enters a **bridge rectifier** (P-N diodes) and a filter capacitor: about 325 V DC.
2. A switching circuit chops this at high frequency into a small high-frequency transformer (step-down and isolation).

3. A fast (Schottky) diode **rectifies** the transformer output again, and a capacitor filters it.
4. Feedback regulates it to **5 V** (USB standard) for the phone.

> ★ KEY POINT
>
> **Function of the rectifier (M-S24, A-S25):** to turn AC into DC, because batteries can only be charged by DC. **Why raw rectifier output cannot charge a phone (M-S26):** it is pulsating, unregulated and at the wrong voltage; USB needs a steady 5 V, so it must be filtered and regulated first.

Regulated 5 V / 3.3 V supply for Arduino and Raspberry Pi

1. Adapter or mains supply → (bridge rectifier + filter) → unregulated DC (Arduino barrel jack: 7-12 V recommended).
2. **Reverse-polarity diode** in series at the input (blocks a wrongly connected adapter).
3. **5 V linear regulator** (e.g. 7805 / NCP1117) gives 5 V; a second **3.3 V regulator** (AMS1117-3.3) feeds 3.3 V parts.
4. Filter capacitors at input and output; a power LED shows it is on.

> № HIGH-YIELD NUMBER
>
> **Arduino Uno operating voltage: 5 V** (A-S28); recommended input 7-12 V. Raspberry Pi: 5 V at about 3 A (USB-C), logic at 3.3 V.

Why a Raspberry Pi needs a stable supply (M-S27) / problems of an inadequate supply (M-L29):

- ► **Voltage drops (under-voltage):** when current demand rises, a weak adapter or thin cable lets the voltage sag below 4.75 V.
- ► **Resets / brown-outs:** the processor restarts or freezes when the voltage dips.
- ► **Data corruption:** a dip while writing can corrupt the SD card.
- ► **Overheating:** a regulator dropping too much voltage, or a struggling adapter, gets hot.
- ► **Unstable operation:** USB devices disconnect, Wi-Fi drops, sensor/ADC readings become noisy, the Pi throttles its speed.

How P-N diodes are used in the Arduino 5 V supply (N-L29): bridge rectifier in the adapter; series diode for reverse-polarity protection; protection diodes on the regulator; Zener or TVS diodes for over-voltage.

6.4 LED indicators and protection diodes

QB: M-L31, A-L32, A-S30, N-S24, N-S25

LED indicator circuit (M-L31, A-L32): supply → resistor → LED (anode to +) → ground, often switched by a transistor or a microcontroller pin. Polarity matters (long leg +); V_F 1.8-3.3 V; $R = (V_S - V_F)/I_F$.

- ► **Uses:** power-on lamp, charging status (red = charging, green = full), standby, network activity, fault warning.
- ► **Examples (A-S30):** phone charger, laptop, Wi-Fi router, TV standby light, washing machine, power bank.

▣ **DEFINITIONS**

Reverse-polarity protection (N-S25): a diode in series with the supply conducts only when the supply is connected the right way; if it is reversed, the diode is reverse biased and blocks current, so the circuit is not damaged. (Alternative: a reverse diode across the supply with a fuse.)

Protection diodes on Arduino/Pi pins (N-S24): clamp diodes from each I/O pin to V_{CC} and to ground conduct if the pin voltage goes above V_{CC} + 0.7 V or below −0.7 V, steering ESD and over-voltage away from the chip. A **flyback diode** across a relay or motor coil absorbs its back-EMF spike.

TEST YOURSELF

Close the notes and answer first. Answers at the end of the chapter.

6.4-1 RECALL Give two products that use LED indicators and what they show.

6.4-2 APPLY Why does a series diode cut the battery voltage reaching the circuit?

ANSWERS — COVER UNTIL YOU HAVE TRIED

APPLY **6.2-1** (5 − 3.2)/0.015 = 120 Ω.

NEW SETTING **6.2-2** R = 10/0.02 = 500 Ω (use 560 Ω); P = 0.02^2 × 500 = 0.2 W, so use a 1/2 W resistor.

RECALL **6.4-1** Phone charger (charging / full), router (network activity), TV (standby).

APPLY **6.4-2** It drops about 0.7 V when conducting (use a Schottky, 0.3 V, to lose less).

★ Formula sheet and numbers

Every formula in the unit

Topic	Formula	Notes
Thermal voltage	$V_T = kT/q$	T in K; 25.9 mV at 300 K
Diode equation	$I_D = I_s(e^{V_D/nV_T} - 1)$	$V = nV_T \ln(I/I_s + 1)$
Temperature	$I_s(T) = I_s \times 2^{\Delta T/10}$	knee −2.5 mV/°C
Dynamic resistance	$r_d = \Delta V/\Delta I \approx 26\ \text{mV}/I_D$	static R = V/I
Transformer	$V_s = V_p N_2/N_1$, $V_m = \sqrt{2}\, V_{rms}$	230 V is RMS
HWR	$V_{dc} = V_m/\pi$, $I_{dc} = I_m/\pi$, $I_{rms} = I_m/2$	γ 1.21, η 40.6%, PIV V_m
FWR (CT, bridge)	$V_{dc} = 2V_m/\pi$, $I_{rms} = I_m/\sqrt{2}$	γ 0.482, η 81.2%
Peak current	$I_m = V_m/(r_f + R_L)$	
Ripple factor	$\gamma = \sqrt{((I_{rms}/I_{dc})^2 - 1)}$	
Efficiency	$\eta = I_{dc}^2 R_L / I_{rms}^2 (r_f + R_L)$	
Regulation	$(V_{NL} - V_{FL})/V_{FL} \times 100$	ideal 0%
C-filter ripple	$V_{r(p\text{-}p)} = V_m/(2fCR_L)$ full-wave; $/(fCR_L)$ half-wave	$V_{dc} = V_m - V_r/2$
Zener	$I_S = (V_{in} - V_Z)/R_S$, $I_Z = I_S - I_L$, $P_Z = V_Z I_Z$	test: $V_{in}R_L/(R + R_L) \geq V_Z$
Zener design	$R = (V_{in(min)} - V_Z)/(I_{L(max)} + I_{Z(min)})$	$I_{ZM} = P_{ZM}/V_Z$
LED resistor	$R = (V_S - V_F)/I_F$, $P = I^2R$	

High-yield numbers

0.7 V / 0.3 V / 1.2 V
Knee / barrier of Si / Ge / GaAs

1.12 eV / 0.72 eV
Band gap of Si / Ge

25.9 mV
V_T at 27 °C (300 K); ≈ 26 mV

×2 per 10 °C
Rise of I_s with temperature

−2.5 mV/°C
Shift of forward voltage with temperature

0.318 / 0.636
V_{dc}/V_m for HWR / FWR

1.21 / 0.482
Ripple factor HWR / FWR

40.6% / 81.2%
Max efficiency HWR / FWR

0.287 / 0.693 / 0.812
TUF: HWR / centre-tap / bridge

V_m / $2V_m$ / V_m
PIV: HWR / CT / bridge

< 5 V / > 6 V
Zener / avalanche breakdown

5 V
Arduino Uno operating voltage; USB standard

1.8-2.2 V
Red LED forward voltage

100 Hz
Full-wave ripple frequency on 50 Hz mains

Glossary (quotable definitions)

Semiconductor
Material with conductivity between conductor and insulator, controlled by temperature, light and impurities.

Doping
Adding a controlled impurity to a pure semiconductor to raise its conductivity.

Depletion layer
Region at a P-N junction emptied of free carriers, holding only fixed ions.

Barrier potential
Potential across the depletion layer that opposes majority-carrier diffusion (Si 0.7 V, Ge 0.3 V).

Knee voltage
Forward voltage at which a diode starts to conduct significantly.

Breakdown voltage
Reverse voltage at which reverse current rises sharply.

Reverse saturation current
Small reverse current carried by minority carriers; set by temperature, not voltage.

Rectifier
Circuit that converts AC into pulsating DC.

PIV
Maximum reverse voltage across a diode when it is not conducting.

Ripple factor
RMS value of the AC component divided by the DC value of the output.

Rectification efficiency
DC output power divided by AC input power.

TUF
DC power to the load divided by the VA rating of the transformer secondary.

Filter
Circuit that removes ripple from rectified DC.

Voltage regulator
Circuit that keeps output DC constant despite changes of input or load.

Zener diode
Heavily doped diode designed to work in reverse breakdown at a nearly constant V_Z.

Derating factor

Reduction of a device's power rating per °C above 25 °C.

Electroluminescence

Emission of light when electrons and holes recombine in a forward-biased junction.

LED

P-N diode that emits light by electroluminescence when forward biased.

Exam-answer templates for the long questions

HOW TO WRITE IT IN THE EXAM — ANY "DRAW AND EXPLAIN" RECTIFIER

1. Definition of rectifier (1 line).
2. Circuit diagram, labelled.
3. + half-cycle: which diode(s) ON, current path.
4. − half-cycle: which diode(s) ON, current path.
5. Input and output waveforms, one above the other.
6. Formulas: V_{dc}, γ, η, PIV, f_{out}.

HOW TO WRITE IT IN THE EXAM — ZENER AS A VOLTAGE REGULATOR

1. Define regulator and why it is needed.
2. Circuit: R_S in series, Zener reverse biased across R_L.
3. $I_S = I_Z + I_L$, $V_L = V_Z$.
4. The four cases (Table 5.1).
5. Condition: $I_{ZK} < I_Z < I_{ZM}$; one advantage, one limitation.

HOW TO WRITE IT IN THE EXAM — BREAKDOWN MECHANISMS

1. Define breakdown.
2. Zener: heavy doping → thin layer → strong field → tunnelling.
3. Avalanche: light doping → wide layer → collisions → chain reaction.
4. Comparison table with temperature coefficient.
5. Zener V-I sketch.

HOW TO WRITE IT IN THE EXAM — LED

1. Definition + electroluminescence.
2. Construction sketch (Fig 6.2) and symbol.
3. Working in 4 steps.
4. Colour / material table.
5. Advantages and applications; series resistor formula.

HOW TO WRITE IT IN THE EXAM — REGULATED POWER SUPPLY / ARDUINO SUPPLY

1. Block diagram with the waveform under each block.
2. One line per block.
3. For Arduino/Pi: 5 V and 3.3 V regulators, reverse-polarity diode.
4. Problems if inadequate: drops, resets, overheating, instability.

Mixed quick-fire questions (cover the right column)

#	Question	Answer (cover this column)
1	**Majority carrier in N-type?**	Electrons.
2	**Barrier of Ge?**	0.3 V.
3	**Which bias widens the depletion layer?**	Reverse bias.

#	Question	Answer (cover this column)
4	**Name the current in reverse bias.**	Reverse saturation (leakage) current.
5	**Which capacitance dominates in forward bias?**	Diffusion capacitance.
6	**PIV of centre-tap rectifier?**	$2V_m$.
7	**Ripple percentage in HWR?**	121%.
8	**Output frequency of bridge on 50 Hz?**	100 Hz.
9	**Which full-wave rectifier has the highest TUF?**	Bridge (0.812).
10	**How is a filter capacitor connected?**	In parallel with the load.
11	**Ideal regulation value?**	0%.
12	**Temperature coefficient of avalanche breakdown?**	Positive.
13	**Zener breakdown mechanism?**	Quantum tunnelling.
14	**Minimum current to keep a Zener regulating?**	Knee current I_{ZK}.
15	**Working principle of LED?**	Electroluminescence.
16	**Why GaAs, not Si, for LEDs?**	Direct band gap: recombination gives light, not heat.
17	**Purpose of voltage regulator?**	Keep output voltage constant.
18	**Arduino Uno operating voltage?**	5 V.
19	**Regulated vs unregulated supply?**	Regulated output stays fixed when mains or load change; unregulated does not.
20	**Ratio of FWR to HWR efficiency?**	2.
21	**V_{dc} of 230 V, 10 : 1, HWR?**	10.35 V.

Question-bank coverage map

Every question in the three QB-3 banks and where its answer is. M = Maurya, A = Shukla, N = Sonam Singh; L = long, S = short. Page numbers are live.

Q	§	p.	Q	§	p.	Q	§	p.	Q	§	p.
M-L1	§1.5	22	**M-S14**	§3.1	33	**A-L30**	§6.2	56	**N-L13**	§3.7	38
M-L2	§1.4	21	**M-S15**	§3.3	35	**A-L31**	§6.3	57	**N-L14**	§3.7	38
M-L3	§1.3	20	**M-S16**	§5.1	48	**A-L32**	§6.4	58	**N-L15**	§3.5	36
M-L4	§1.5	22	**M-S17**	§5.2	49	**A-S1**	§1.2	19	**N-L16**	§4.2	45

Q	§	p.	Q	§	p.	Q	§	p.	Q	§	p.
M-L5	§2.1	29	**M-S18**	§5.2	49	**A-S2**	§1.4	21	**N-L17**	§3.7	38
M-L6	§2.1	29	**M-S19**	§5.2	49	**A-S3**	§3.3	35	**N-L18**	§3.4	35
M-L7	§1.7	26	**M-S20**	§5.2	49	**A-S4**	§1.2	19	**N-L19**	§5.3	50
M-L8	§1.6	23	**M-S21**	§5.1	48	**A-S5**	§1.2	19	**N-L20**	§2.1	29
M-L9	§1.6	23	**M-S22**	§6.1	55	**A-S6**	§1.2	19	**N-L21**	§5.4	51
M-L10	§3.2	33	**M-S23**	§6.2	56	**A-S7**	§1.1	18	**N-L22**	§5.4	51
M-L11	§3.7	38	**M-S24**	§6.3	57	**A-S8**	§1.2	19	**N-L23**	§5.3	50
M-L12	§4.2	45	**M-S25**	§4.1	44	**A-S9**	§1.1	18	**N-L24**	§5.4	51
M-L13	§3.5	36	**M-S26**	§6.3	57	**A-S10**	§1.7	26	**N-L25**	§5.1	48
M-L14	§3.4	35	**M-S27**	§6.3	57	**A-S11**	§3.6	37	**N-L26**	§5.4	51
M-L15	§3.7	38	**M-S28**	§6.2	56	**A-S12**	§3.3	35	**N-L27**	§6.1	55
M-L16	§3.2	33	**A-L1**	§1.3	20	**A-S13**	§3.3	35	**N-L28**	§6.3	57
M-L17	§3.7	38	**A-L2**	§1.4	21	**A-S14**	§4.1	44	**N-L29**	§6.3	57
M-L18	§3.6	37	**A-L3**	§1.4	21	**A-S15**	§3.3	35	**N-L30**	§6.2	56
M-L19	§4.1	44	**A-L4**	§1.8	27	**A-S16**	§3.3	35	**N-S1**	§1.3	20
M-L20	§5.1	48	**A-L5**	§1.7	26	**A-S17**	§3.5	36	**N-S2**	§1.2	19
M-L21	§5.4	51	**A-L6**	§2.1	29	**A-S19**	§5.1	48	**N-S3**	§1.3	20
M-L22	§5.3	50	**A-L7**	§1.6	23	**A-S20**	§6.2	56	**N-S4**	§1.5	22
M-L23	§5.4	51	**A-L8**	§1.6	23	**A-S21**	§5.2	49	**N-S5**	§1.3	20
M-L24	§5.4	51	**A-L9**	§1.6	23	**A-S22**	§5.2	49	**N-S6**	§1.5	22
M-L25	§5.3	50	**A-L10**	§3.5	36	**A-S23**	§6.1	55	**N-S7**	§1.4	21
M-L26	§6.1	55	**A-L11**	§3.7	38	**A-S24**	§6.1	55	**N-S8**	§1.4	21
M-L27	§6.2	56	**A-L12**	§4.2	45	**A-S25**	§6.3	57	**N-S9**	§1.2	19
M-L28	§4.3	46	**A-L13**	§3.2	33	**A-S26**	§4.3	46	**N-S10**	§1.4	21
M-L29	§6.3	57	**A-L14**	§3.4	35	**A-S27**	§4.3	46	**N-S11**	§3.3	35
M-L30	§4.3	46	**A-L15**	§3.7	38	**A-S28**	§6.3	57	**N-S12**	§3.3	35
M-L31	§6.2	56	**A-L16**	§3.5	36	**A-S29**	§4.3	46	**N-S13**	§3.3	35
M-S1	§1.2	19	**A-L17**	§3.7	38	**A-S30**	§6.4	58	**N-S14**	§1.6	23
M-S2	§6.1	55	**A-L18**	§3.8	41	**N-L1**	§1.3	20	**N-S15**	§3.3	35
M-S3	§1.4	21	**A-L19**	§4.1	44	**N-L2**	§1.7	26	**N-S16**	§3.6	37
M-S4	§1.10	28	**A-L20**	§3.5	36	**N-L3**	§1.8	27	**N-S17**	§5.1	48
M-S5	§1.1	18	**A-L21**	§5.1	48	**N-L4**	§1.8	27	**N-S18**	§5.2	49
M-S6	§1.6	23	**A-L22**	§5.1	48	**N-L5**	§1.10	28	**N-S19**	§5.2	49
M-S7	§1.1	18	**A-L23**	§5.1	48	**N-L6**	§1.6	23	**N-S20**	§5.3	50
M-S8	§1.1	18	**A-L24**	§5.3	50	**N-L7**	§1.4	21	**N-S21**	§6.1	55
M-S9	§1.5	22	**A-L25**	§5.4	51	**N-L8**	§3.1	33	**N-S22**	§6.1	55
M-S10	§3.6	37	**A-L26**	§5.4	51	**N-L9**	§3.2	33	**N-S23**	§6.2	56
M-S11	§3.3	35	**A-L27**	§5.3	50	**N-L10**	§3.5	36	**N-S24**	§6.4	58
M-S12	§3.1	33	**A-L28**	§4.3	46	**N-L11**	§4.1	44	**N-S25**	§6.4	58
M-S13	§3.3	35	**A-L29**	§6.1	55	**N-L12**	§3.7	38			

How to revise with these notes

Each time you sit down to revise:

1. **Start closed-book.** Answer a chapter's Retrieve boxes and practice *before* rereading anything. Rereading feels productive; recalling is what makes it stick.

2. **Then check.** Mark against the key. Reread only what you missed or got only half right.
3. **Come back more than once.** Several shorter sessions with gaps between them beat one long one. When is up to you.
4. **Mix it up later.** On later passes, mix chapters (the mixed Q&A or mixed practice) so you practise telling topics apart.
5. **Finish on the summary.** The one-page summary is for the last look, not for first learning.

Exam within a day or two and the unit is new to you? Use the Pass Kit at the front instead.

■ One-page summary

Semiconductors

Si 1.12 eV, Ge 0.72 eV. Doping: N (P, As, Sb; electrons) and P (B, Al, Ga, In; holes).

P-N junction

Diffusion → recombination → ions → depletion layer → barrier 0.7 / 0.3 V. FB narrows, RB widens.

V-I and equation

Knee 0.7 V; tiny I_s; breakdown V_{BR}. $I = I_s(e^{V/nV_T} - 1)$, $V_T = kT/q$.

Temperature

Curve shifts left 2.5 mV/°C; I_S doubles per 10 °C.

Diode circuits

ON → 0.7 V source; OFF → open. Parallel: lowest drop wins.

Rectifiers

HWR V_m/π, 1.21, 40.6%, PIV V_m. CT and bridge $2V_m/\pi$, 0.482, 81.2%; PIV $2V_m$ / V_m.

Filters

C across load charges at peaks. $V_r = V_m/(2fCR_L)$. Supply: transformer, rectifier, filter, regulator.

Zener

Tunnelling (− tc, < 5 V) vs avalanche (+ tc, > 6 V). Shunt regulator: test, then $I_S = I_Z + I_L$.

LED

Electroluminescence; GaAs/GaN; $R = (V_S - V_F)/I_F$.

Applications

USB charger: rectify, filter, regulate to 5 V. Arduino 5 V / 3.3 V regulators; reverse-polarity diode; clamp diodes.

LAST LINE OF DEFENCE

A diode is a one-way valve with a 0.7 V toll. Rectifiers use it to turn AC into DC, capacitors smooth it, a reverse-biased Zener holds it steady, and an LED turns the same junction into light.